

VISVESVARAYA TECHNOLOGICAL UNIVERSITY

“Jnana Sangama”, Belagavi – 590018, Karnataka State, India



A Project Report on

**“A Hybrid Neural Network for EEG-based Depression
Screening Using CNN+LSTM”**

Submitted in partial fulfillment of requirements for the award of the degree

BACHELOR OF ENGINEERING

IN

Artificial Intelligence & Data Science

Submitted by:

MAREDDY CHARAN 1SJ22AD024

Under the guidance of:

Mr. Badrinath K

Assistant Professor

Dept. of AI&DS, SJCIT



**DEPARTMENT OF ARTIFICIAL INTELLIGENCE & DATA SCIENCE
SRI JAGADAGURU CHANDRASHEKARANATHA SWAMIJI
INSTITUTE OF TECHNOLOGY**

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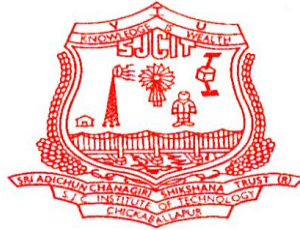
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**SRI JAGADGURU CHANDRASHEKARANATHA SWAMIJI INSTITUTE OF
TECHNOLOGY,**

CHICKBALLAPUR- 562 101 (KARNATAKA)

DEPARTMENT OF ARTIFICIAL INTELLIGENCE & DATA SCIENCE



CERTIFICATE

This is to certify that the Project work entitled “**A Hybrid Neural Network for EEG-based Depression Screening Using CNN+LSTM**” presented is a bonafied work carried out by Mr. **MAREDDY CHARAN**, bearing register number **1SJ22AD024**, in partial fulfillment for the award of **Bachelor of Engineering in Artificial Intelligence & Data Science**, from **Visvesvaraya Technological University, Belagavi** during the year of **2025-26**. It is certified that all corrections / suggestions indicated for internal assessment have been incorporated in the report. This project report has been approved as is satisfies the academic requirements in respect of project work prescribed for the said degree.

Signature of the Guide

Prof. Badrinath K

Signature of the HOD

Dr. Nagaraja G

Signature of the Principal

Dr. G T Raju

External Examiners:

Name of the Examiners:

Signature with Date

1. -----

2. -----

DECLARATION

I declare that this project report titled “**A Hybrid Neural Network for EEG-based Depression Screening Using CNN+LSTM**” submitted in partial fulfillment of the degree of B. Tech in **Department of Artificial Intelligence and Data Science** is a record of original work carried out by me under the supervision of **Mr. Badrinath K** and has not formed the basis for the award of any other degree, in this or any other Institution or University. In keeping with the ethical practice in reporting scientific information, due acknowledgements have been made wherever the findings of others have been cited.

Signature of the student

Name

MAREDDY CHARAN

Signature

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With reverential pranam, I express my sincere gratitude and salutations to the feet of his holiness **Paramapoojya Jagadguru Byravaikya Padmabhushana Sri Sri Sri Dr. Balagangadharanatha Maha Swamiji**, his holiness **Parama poojya Jagadguru Sri Sri Sri Dr. Nirmalanandanatha Maha Swamiji**, and **Sri Sri Mangalnath Swamiji**, Sri Adichunchanagiri Mutt for their unlimited blessings.

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ABSTRACT

Depression is a psychological disorder characterized by the continuous occurrence of a low mood state. It significantly affects individuals across multiple age groups worldwide and has become a serious global health concern. Early diagnosis plays a vital role in preventing its long-term impact and can potentially save lives. Electroencephalogram (EEG) signals serve as a reliable diagnostic tool, as the analysis of brain wave patterns can reflect the underlying mental state of an individual.

A fully automated depression detection system offers advantages over manual EEG analysis, which is time-consuming, labor-intensive, and requires significant expertise. A novel computer-aided hybrid neural network model, known as DepHNN (Depression Hybrid Neural Network), has been developed for depression screening using EEG signals. The model integrates a Convolutional Neural Network (CNN) for spatial and temporal feature extraction, along with Long Short-Term Memory (LSTM) layers for learning sequential patterns. The use of a windowing technique reduces computation complexity and training time. EEG data was collected from 21 drug-free symptomatic depressed patients and 24 normal individuals using a neuroscan system. The model achieved a high classification accuracy of 99.10% and a mean absolute error (MAE) of 0.2040, demonstrating its effectiveness, low complexity, and strong potential for real-time depression detection.

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CHAPTER 1

INTRODUCTION

CHAPTER 1

INTRODUCTION

1.1 OVERVIEW

Depression has emerged as one of the most critical public health challenges of the modern era, affecting individuals' emotional, cognitive, and physical functioning. Globally, more than 322 million people are estimated to suffer from depression, making it one of the leading causes of disability and reduced quality of life. The impact of depression extends beyond the affected individual, influencing families, workplaces, and society through increased healthcare costs, reduced productivity, and social withdrawal. Although depression is treatable, many cases remain undiagnosed or untreated due to social stigma, lack of awareness, inconsistent diagnostic practices, and limited access to mental health services.

Traditional diagnostic methods rely on psychological evaluations, structured clinical interviews, and self-report assessments. These methods, while widely practiced, are subjective and depend heavily on the patient's ability to articulate their feelings and the clinician's interpretation. This dependence on subjective input can lead to delayed or inaccurate diagnosis, particularly among individuals who underreport symptoms. This situation highlights the need for objective, reliable, and automated diagnostic systems capable of supporting clinical decision-making and enhancing the accuracy and consistency of depression assessment.

Electroencephalography (EEG) offers a non-invasive means of capturing the brain's electrical activity and has shown value in identifying neural irregularities associated with depressive disorders[1]. EEG signals contain distinct oscillatory patterns and connectivity characteristics that differ between healthy and depressed individuals. However, the interpretation of EEG data is complex due to its non-linear and dynamic nature, requiring specialized expertise and extensive preprocessing. Traditional machine learning methods often rely on handcrafted feature extraction, which is time-consuming and prone to inconsistencies.

The MDD Patients and Healthy Controls EEG Data (New) dataset from Figshare is utilized, containing EEG recordings from individuals clinically diagnosed with Major Depressive Disorder (MDD) and from healthy control participants[1]. The dataset consists of

multi-channel EEG signals collected under standardized resting-state conditions, typically with eyes closed, ensuring stable and consistent neural activity capture. All participants were medically evaluated prior to data collection, which supports the reliability and accuracy of the recorded brainwave patterns. The availability of data from both depressive and healthy subjects allows for balanced training and unbiased classification performance.

Before model processing, the EEG signals undergo a series of pre-processing steps including band-pass filtering, artifact removal (such as noise from eye blinks and muscle movement), time-window segmentation, and normalization to enhance signal clarity and prepare the data for deep learning analysis. These refined EEG segments are then used for feature extraction and temporal pattern learning within the CNN–LSTM architecture for accurate mental state classification.

To address the challenges of EEG interpretation, We proposes a Hybrid Neural Network Model (DepHNN) that incorporates both Convolutional Neural Networks (CNN) and Long Short-Term Memory (LSTM) networks[4]. The CNN component is responsible for extracting salient spatial and temporal features from EEG signals, while the LSTM component captures sequential patterns and long-term dependencies present in time-series neural data[11]. This hybrid architecture allows the model to learn and classify EEG patterns directly from raw signal input, eliminating the need for manual feature engineering.

Experimental results demonstrate that the DepHNN model achieves high classification accuracy with reduced error and computational efficiency compared to several existing diagnostic approaches. The proposed system offers a reliable, fast, and automated solution for depression screening and has the potential to assist clinicians in early detection and intervention. The model is suitable for integration into cloud-based assessment platforms, portable EEG monitoring devices, and remote mental health evaluation systems, supporting broader accessibility.

The integration of the DepHNN model into clinical and digital health environments contributes to the development of more objective and data-driven mental healthcare practices. The system provides consistent diagnostic support, reduces subjectivity in evaluation, and enables screening in regions where access to trained mental health professionals is limited. The approach also establishes a foundation for future extensions, including continuous mood monitoring, depression severity estimation, and treatment response tracking. The work aims

to advance mental healthcare toward more precise, proactive, and personalized practices, supporting improved outcomes for individuals affected by depressive disorders.

1.2 PROBLEM STATEMENT

Depression is a complex and debilitating mental health disorder that significantly affects emotional stability, cognitive processing, behavior, and overall quality of life. It is recognized internationally as one of the leading causes of disability and reduced daily functioning. Although multiple treatment options are available, a substantial portion of individuals do not receive timely diagnosis or care. This is due to several barriers, including social stigma, hesitation to seek psychological help, lack of awareness, and insufficient mental health infrastructure in many communities. The consequences of undiagnosed or untreated depression can be severe, leading to chronic emotional distress, social isolation, reduced work productivity, academic decline, and in extreme cases, suicidal ideation or self-harm.

Current clinical procedures for diagnosing depression involve structured interviews, clinical observations, and standardized rating scales such as the Beck Depression Inventory (BDI) and Hamilton Depression Rating Scale (HDRS). These tools rely on subjective self-reporting and the clinician's interpretation of symptoms. Such subjectivity introduces a high degree of variability and inconsistency, particularly when patients struggle to express their feelings accurately or consciously suppress emotional distress. This is due to several barriers, including social stigma, hesitation to seek psychological help, lack of awareness, and insufficient mental health infrastructure in many communities. Additionally, the symptoms of depression can overlap with anxiety disorders, bipolar disorders, or neurological conditions, leading to misdiagnosis or delayed identification. This lack of objective diagnostic measures represents a major limitation in contemporary mental health care practices.

Electroencephalography (EEG) provides a method for capturing the electrical activity of the brain and offers potential as an objective biomarker for diagnosing psychiatric conditions [13]. Research indicates that individuals experiencing depression exhibit measurable differences in neural oscillations, functional connectivity patterns, and cortical activation levels when compared to non-depressed individuals. However, EEG data is inherently complex, high-dimensional, and prone to noise generated from blinking, muscle movement, and environmental interference. Manual interpretation requires specialized expertise and is vulnerable to human error. Traditional machine learning approaches have

attempted to classify depression using EEG signals, but they depend heavily on handcrafted feature extraction. This process is time-consuming, domain-specific, and may fail to capture subtle but important signal patterns.

The key challenge lies in developing an automated, robust, and scalable diagnostic approach capable of analyzing EEG signals without extensive manual preprocessing. DepHNN Analyzer aims to address this challenge by constructing a hybrid deep learning system that integrates Convolutional Neural Networks (CNN) for automatic spatial-temporal feature extraction and Long Short-Term Memory (LSTM) networks for identifying sequential dependencies and long-range neural signal behaviors[11]. By enabling the model to learn representations directly from raw EEG signals, the system reduces reliance on human interpretation and handcrafted features. The goal is to build a reliable depression screening framework that can differentiate between depressed and non-depressed individuals with high accuracy, minimized error, and computational efficiency.

This system is intended to function as a clinical decision-support tool that assists psychologists and psychiatrists in making informed diagnostic judgments. It also holds potential for integration into remote mental health platforms and portable EEG monitoring devices, making it suitable for both clinical and community-based screening. The development of such a model contributes toward a more objective, data-driven approach in the mental healthcare domain, supporting early intervention and improving overall treatment outcomes for individuals affected by depressive disorders.

A system of this nature is particularly valuable in the current global context, where demand for mental health services continues to rise while availability of trained mental health professionals remains limited. By providing fast and consistent assessments, the proposed model can help bridge the diagnostic gap in primary healthcare settings, rural regions, academic institutions, and workplace wellness programs. In addition, its ability to analyze EEG signals in real time offers the possibility of continuous mood monitoring, allowing clinicians to track changes in patient condition and optimize treatment strategies. As advancements continue in deep learning, neural signal processing, and wearable sensor technology, solutions such as the proposed hybrid network have the potential to transform mental health diagnostics into a proactive, scalable, and technology-supported domain. The goal is to build a reliable depression screening framework that can differentiate between depressed and non-depressed individuals with high accuracy, minimized error, and computational efficiency.

1.3 PROBLEM DEFINATION

Depression is a complex psychiatric disorder that significantly affects emotional stability, cognitive performance, and social functioning. The diagnosis of depression in current clinical practice depends largely on subjective evaluation methods, including self-reported questionnaires, therapist-led interviews, and behavioral interpretation scales. These methods rely on an individual's willingness and ability to describe internal emotional experiences, which may be influenced by personal perception, cultural expression, social stigma, or emotional suppression. As a result, diagnostic outcomes can vary greatly between practitioners and settings, often leading to delayed detection, misdiagnosis, or underdiagnosis. This lack of diagnostic objectivity remains a major barrier to early intervention and effective treatment planning.

Neurophysiological indicators such as electroencephalography (EEG) have shown potential for providing objective measurement of brain activity associated with depression. However, EEG signals are inherently complex, nonlinear, and prone to noise artifacts, making manual analysis difficult and time-consuming. Existing machine learning techniques used to classify EEG patterns commonly depend on handcrafted feature extraction steps, which require expert domain knowledge and may overlook subtle neural characteristics. These limitations reduce model generalization and hinder widespread clinical adoption.

There is a clear need for an automated, robust, and scalable system capable of analyzing EEG signals without extensive manual intervention. The challenge lies in designing a computational framework that can effectively learn both spatial features (related to frequency and activation patterns) and temporal characteristics (related to sequential neural behavior) from EEG data. To address this challenge, a hybrid deep learning architecture combining Convolutional Neural Networks (CNN) and Long Short-Term Memory (LSTM) networks is proposed [11]. This integrated approach allows the system to automatically extract meaningful patterns from EEG signals and identify depression-linked neural activity more accurately and consistently.

The goal is to create a clinically relevant depression screening model that reduces reliance on subjective evaluation, enhances diagnostic precision, and supports timely intervention. By enabling objective, data-driven assessment, the system aims to contribute to accessible and technology-assisted mental healthcare that can be applied in clinical, community, and remote monitoring environments.

1.4 PURPOSE

An automated depression screening system using EEG signals and deep learning techniques is designed to provide a more objective, accurate, and reliable approach to mental health diagnosis. Traditional diagnostic methods rely heavily on subjective self-reports and clinical evaluations, which can be influenced by communication barriers, emotional suppression, social stigma, and variations in clinician interpretation. These limitations often result in delayed diagnosis or inappropriate treatment, affecting the overall well-being of individuals.

A hybrid deep learning model that integrates Convolutional Neural Networks (CNN) and Long Short-Term Memory (LSTM) networks is implemented to overcome these challenges[12]. CNN layers extract spatial and temporal features from EEG recordings, while LSTM layers capture sequential dependencies and long-term patterns in brain signal activity. This architecture allows the system to process raw EEG data without manual feature extraction, reducing human error and improving diagnostic consistency.

The system functions as a diagnostic support tool that provides data-driven insights to assist mental health professionals in making informed clinical decisions. Its adaptable and scalable design allows deployment in telemedicine platforms, clinical environments, and portable EEG monitoring setups, enabling remote mental health assessment in regions with limited medical resources. This approach promotes early identification of depressive symptoms, enhances diagnostic accuracy, and supports timely therapeutic intervention.

By integrating intelligent computational techniques with EEG-based analysis, the model contributes to the advancement of proactive, precise, and accessible mental healthcare, improving the quality of life and treatment outcomes for individuals affected by depressive disorders.

1.5 SCOPE

EEG signals obtained from clinically validated depressed individuals and healthy volunteers form the basis for system development and evaluation. The dataset undergoes preprocessing steps such as artifact removal, filtering, segmentation, and normalization to ensure signal clarity and enhance model reliability. A hybrid deep learning architecture integrating CNN and LSTM is trained on these refined EEG segments to recognize distinguishing neural activity patterns associated with depressive states.

The developed system is designed to function not only in controlled laboratory environments but also in real-time and remote assessment settings through integration with portable EEG acquisition devices. The design supports seamless incorporation into clinical diagnostic workflows, primary healthcare screening, university mental health services, and telemedicine platforms. Automated processing and classification reduce reliance on manual EEG interpretation and subjective psychological evaluation, supporting consistent diagnostic decision-making.

The framework emphasizes efficiency, adaptability, and scalability, allowing for performance optimization across different EEG datasets and recording conditions. It enables potential extension toward continuous emotional state monitoring, assessment of treatment response over time, and identification of early indicators of relapse. Future enhancements may include multi-modal data fusion, personalized brain activity profiling, and on-device deployment for wearable neuro-monitoring systems. The intention is to support accessible, objective, and data-driven mental health evaluation across diverse healthcare environments.

The approach aligns with the growing movement toward technologically assisted mental healthcare, where biological signals and machine intelligence complement psychological assessment. By leveraging neural activity patterns rather than self-reported mood descriptions alone, the system contributes to more objective diagnostic pathways. This shift supports early identification, reduces the chances of misclassification, and provides a foundation for targeted therapeutic planning. The resulting model encourages a transition from reactive treatment to proactive mental health management, ultimately benefiting individuals, caregivers, and mental health practitioners.

1.6 OBJECTIVES

1. Design a CNN-LSTM model to automate depression detection using EEG signals.
2. Improve adaptability using cross-validation and diverse datasets while employing regularization techniques like L2 regularization and adaptive learning rates.
3. Enhance model accuracy and generalization through techniques such as dropout, data augmentation, and early stopping.
4. Reduce computational complexity and latency for real-time depression screening on portable and cloud-based platforms.

5. Pre-process EEG signals effectively by applying filtering, artifact removal, and normalization to ensure high-quality input for the deep learning model.

1.7 ORGANIZATION OF CHAPTERS

Chapter 1:- Introduction

Chapter 2:- Literature Survey

Chapter 3:- System Analysis

Chapter 4:- Software Specification Requirements

Chapter 5:- System Design

Chapter 6:- Implementation

Chapter 7:- Testing

Chapter 8:- Conclusion

Bibliography

Appendix A: - Snapshots

CHAPTER 2
LITERATURE SURVEY

CHAPTER 2

LITERATURE SURVEY

- **Acharya, U. R., Oh, S. L., Hagiwara, Y., Tan, J. H., & Adeli, H. (2018). Deep convolutional neural network for the automated detection and diagnosis of seizure using EEG signals.[2]**

An encephalogram (EEG) is a commonly used ancillary test to aid in the diagnosis of epilepsy. The EEG signal contains information about the electrical activity of the brain. Traditionally, neurologists employ direct visual inspection to identify epileptiform abnormalities. This technique can be time-consuming, limited by technical artifact, provides variable results secondary to reader expertise level, and is limited in identifying abnormalities. Therefore, it is essential to develop a computer-aided diagnosis (CAD) system to automatically distinguish the class of these EEG signals using machine learning techniques. This is the first study to employ the convolutional neural network (CNN) for analysis of EEG signals. In this work, a 13-layer deep convolutional neural network (CNN) algorithm is implemented to detect normal, preictal, and seizure classes. The proposed technique achieved an accuracy, specificity, and sensitivity of 88.67%, 90.00% and 95.00%, respectively.[2]

Advantages:

- Introduces a deep convolutional neural network (CNN) for automated seizure detection using EEG signals, reducing the need for manual interpretation.
- Demonstrates high accuracy in detecting seizures, showcasing the potential of deep learning in medical diagnostics.

Disadvantages:

- The model's performance may vary with different EEG datasets, indicating potential challenges in generalizability.
- Deep learning models require substantial computational resources and large labelled datasets for training.

- **Rathi, S., Patel, V., & Agrawal, R. (2023). [4] Deep learning and machine learning in psychiatry: A survey of current progress in depression detection, diagnosis, and treatment.**

Recent advances in brain and mental health research are driven by technologies like machine learning, deep learning, and AI, enabling more personalized and precise approaches in diagnosing and treating depression. [4] Precision psychiatry uses these tools to tailor care, identify at-risk individuals, and challenge traditional diagnostic categories through unsupervised learning.[4] While AI supports all treatment phases and offers evidence-based alternatives, current research lacks empirical proof of better patient outcomes. To advance clinical use, there's a need for rigorous validation, better data access, interdisciplinary collaboration, and standardized definitions.

Advantages:

- Explores the application of deep learning and machine learning in detecting and treating depression, emphasizing precision psychiatry.
- Discusses the potential of AI to provide personalized mental health care.

Disadvantages:

- Notes that many AI models lack empirical validation in clinical settings.
- Highlights challenges like data heterogeneity and the need for standardized definitions in psychiatric research.

- **Saha, S., & Dasgupta, R. (2022). Hybrid CNN-LSTM architecture for EEG-based human activity recognition.[5] In Proceedings of the International Conference on Intelligent Computing and Smart Communication 2022**

Human Activity Recognition (HAR) involves predicting a person's activity using sensor data from wearable devices. We use a hybrid CNN-LSTM model to improve recognition accuracy, especially during activity transitions. [5] CNN handles spatial features from raw data, while LSTM captures temporal transitions. The model is trained on HAPT and HAR datasets, with performance optimized through hyper parameter tuning. Evaluation metrics like precision, recall, and F1-score show improved accuracy with optimized settings.

Advantages:

- Combines CNN and LSTM architectures to capture both spatial and temporal features in EEG data, enhancing human activity recognition.
- Demonstrates improved accuracy over traditional methods.

Disadvantages:

- Hybrid models can be computationally intensive, requiring more resources for training and deployment.
 - May face challenges in real-time applications due to increased complexity.
- **Chen, S., Yu, Y., & Pan, J. (2023). [6] MADNet: EEG-Based Depression Detection Using a Deep Convolution Neural Network Framework with Multi-dimensional Attention. In Artificial Neural Networks and Machine Learning – ICANN 2023[6]**

Major depressive disorder (MDD) is a common and serious mental health problem that has received increasing attention from both researchers and clinicians. Electroencephalography (EEG)-based automatic diagnosis of MDD has been explored in previous studies, but feature extraction remains an area for improvement. We propose a novel deep learning approach for the automated detection of MDD. For feature extraction, a multi-dimensional attention mechanism is used to extract features from a simple pre-processed EEG signal to enhance spatio-temporal feature learning.[6] We randomly divided EEG data from 26 MDD patients and 26 healthy controls into training and testing sets for network training and testing, respectively. Our method achieved the highest accuracy of 95.14% in subject-based data classification experiments compared to other methods. As a result of its simplicity and convenience, the method proposed in this study has the potential to be utilized as a tool for assisting in the daily diagnosis of depression.

Advantages:

- Introduces MADNet, a deep CNN framework with multi-dimensional attention mechanisms for EEG-based depression detection.
- Achieves high accuracy, showcasing the effectiveness of attention mechanisms in enhancing model performance.

Disadvantages:

- The complexity of the model may hinder its interpretability and real-time application.
 - Requires extensive computational resources for training and deployment.
- **Ahmadlou, M., Adeli, H., & Adeli, A. (2012). [7]Fractality analysis of frontal brain in major depressive disorder. *International Journal of Psychophysiology***

While emotional stimuli are known to affect pain perception, this study explored how pain influences emotion processing. Involving 30 participants, researchers measured muscle responses to happy and angry faces during painful and non-painful stimulation. Pain slowed both smiling and frowning responses, particularly reducing incorrect frowning responses to happy faces, but had no effect on reactions to angry faces or emotional ratings.[7] These findings suggest pain selectively alters the processing of happy faces, potentially impacting social interactions during pain.

Advantages:

- Utilizes fractal analysis to study EEG signals in patients with major depressive disorder, offering insights into brain dynamics.
- Provides a novel perspective on EEG signal analysis beyond traditional linear methods.

Disadvantages:

- Fractal analysis techniques may be sensitive to noise and require careful preprocessing.
 - The study's findings may need further validation across diverse populations.
- **Acharya, U. R., Sudarshan, V. K., Adeli, H., Santhosh, J., Koh, J. E., & Adeli, A. (2015).[10] A novel depression diagnosis index using nonlinear features in EEG signals. *European Neurology*.**

Sex chromosome aneuploidies range in incidence from rather common to exceedingly rare and have a variable phenotype. We report 2 patients with sex chromosome aneuploidies who developed severe aplastic anemia requiring treatment. The first patient

had tetrasomy X and presented at 9 years of age, and the second patient had trisomy and presented at 5 years of age. Although aplastic anemia has been associated with other chromosomal abnormalities, sex chromosome abnormalities have not been traditionally considered a risk factor for this condition. A review of the literature reveals that at least one other patient with a sex chromosome aneuploidy (45,X) has suffered from aplastic anemia and that other autosomal chromosomal anomalies have been described.[10] Despite the uncommon nature of each condition, it is possible that the apparent association is coincidental. A better understanding of the genetic causes of aplastic anemia remains important.

Advantages:

- Introduces a novel depression diagnosis index using nonlinear features in EEG signals, enhancing diagnostic accuracy.
- Combines multiple nonlinear metrics to capture complex brain dynamics associated with depression.

Disadvantages:

- The approach may require extensive computational resources for feature extraction and analysis.
 - Clinical applicability needs further exploration through larger studies.
- **Alhagry, S., Fahmy, A. A., & El-Khoribi, R. A. (2017).[17] Emotion recognition based on EEG using LSTM recurrent neural network. International Journal of Advanced Computer Science and Applications.**

Emotion is the most important component in daily interaction between people. Nowadays, it is important to make the computers understand user's emotion who interacts with it in human-computer interaction (HCI) systems. Electroencephalogram (EEG) signals are the main source of emotion in our body.[17] Recently, emotion recognition based on EEG signals have attracted many researchers and many methods were reported. Different types of features were extracted from EEG signals then different types of classifiers were applied to these features. In this paper, a deep learning method is proposed to recognize emotion from raw EEG signals. Long-Short Term Memory

(LSTM) is used to learn features from EEG signals then the dense layer classifies these features into low/high arousal, valence, and liking. DEAP dataset is used to verify this method which gives an average accuracy of 85.65%, 85.45%, and 87.99% with arousal, valence, and liking classes, respectively. The proposed method introduced high average accuracy in comparison with the traditional techniques.

Advantages:

- Utilizes LSTM to capture temporal dependencies in EEG signals effectively.
- Demonstrates promising performance in emotion classification using deep learning.

Disadvantages:

- Relatively small dataset, which may affect generalizability.
- Focused only on emotion detection, not suitable for broader psychiatric diagnosis.

- **Y. Ding, X. Chen, Q. Fu, S. Zhong [9], A Depression Recognition Method for College Students Using Deep Integrated Support Vector Algorithm.” IEEE Access (2020).**

A deep integrated Support Vector Machine (SVM) framework was developed to identify depression levels among college students using EEG signals. The hybrid model combines deep feature extraction with SVM classification to enhance recognition accuracy. Deep learning layers automatically extract high-level EEG features, while the SVM classifier provides precise decision boundaries, leading to improved generalization even with limited data. Experimental evaluation showed that the integrated method achieved better accuracy and robustness than traditional SVM and standalone deep networks. The work emphasizes the potential of combining deep learning and classical machine-learning approaches for effective EEG-based mental-health screening.

Advantages:

- Integration of deep learning and SVM improves classification performance.
- Efficient handling of smaller datasets through enhanced feature extraction.

- Suitable for practical mental-health monitoring among college students.

Disadvantages:

- High computational cost due to hybrid network design.
 - Sensitive to parameter selection and tuning.
 - Generalization limited to the specific demographic studied.
- **M. Bachmann, L. P"oeske, K. Kalev, K. Aarma, A. Lehtmets, P. " O"opik, J. Lass, H. Hinrikus,[10] Methods for classifying depression in single channel EEG using linear and nonlinear signal analysis, Comput. Methods Programs Biomed. 155 (2018)**

Linear and nonlinear signal-analysis techniques were employed to classify depressive and normal conditions from single-channel EEG recordings. The approach extracted both spectral and nonlinear parameters such as power spectral density, fractal dimension, and entropy to capture differences in brain dynamics associated with depression. Support Vector Machine (SVM) classifiers were then applied to the extracted features to separate the two groups. The results demonstrated that even single-channel EEG can provide valuable diagnostic information when analyzed using combined linear and nonlinear measures. This work highlights the potential for developing portable and simplified EEG-based systems for mental-health screening.

Advantages:

- Effective combination of linear and nonlinear EEG features for accurate classification.
- Demonstrates feasibility of depression detection using minimal EEG channels.
- Reduces complexity and cost of data acquisition.

Disadvantages:

- Limited spatial information due to single-channel recording.
- Accuracy slightly lower than multi-channel or deep-learning models.
- Small dataset may restrict generalization of results.

- **S.D. Puthankattil, P.K. Joseph, Classification of EEG signals in normal and depression conditions by ANN using RWE and signal entropy, J. Mech. Med. Biol. 12 (2012) 1240019[9].**

An Artificial Neural Network (ANN) model was utilized to classify EEG signals recorded from normal and depressive subjects using Relative Wavelet Energy (RWE) and signal entropy as distinguishing features. The extracted RWE values captured frequency-domain energy variations across different EEG sub-bands, while entropy represented the randomness and complexity of neural activity[9]. By training the ANN on these combined features, the model effectively differentiated depressive EEG patterns from normal ones. The approach provided an early example of machine learning applied to mental-health signal processing and demonstrated that wavelet-based energy and entropy measures are valuable indicators of depression-related changes in brain activity.

Advantages:

- Combines wavelet energy and entropy features for enhanced diagnostic accuracy.
- Simple ANN model offers fast training and easy implementation.
- Establishes the foundation for later hybrid EEG-based depression detection systems.

Disadvantages:

- Lacks temporal sequence modeling of EEG signals.
- Performance depends on manual feature extraction and preprocessing.
- Accuracy limited compared to modern deep-learning approaches.
- **O. Faust, Y. Hagiwara, T.J. Hong, O.S. Lih, U.R. Acharya, Deep learning for healthcare applications based on physiological signals: A review, Comput. Methods Programs Biomed. 161 (2018) 1–13[8].**

A comprehensive review of deep-learning applications in healthcare was presented, focusing on physiological signals such as EEG, ECG, and EMG. The paper summarized the major architectures—Convolutional Neural Networks (CNN), Recurrent Neural Networks (RNN), and hybrid frameworks—used for analyzing medical time-series data.

Special attention was given to the role of deep learning in automating feature extraction, improving diagnostic accuracy, and enabling real-time disease detection. The review highlighted how deep models surpass traditional machine-learning techniques in handling complex biomedical patterns, while also addressing challenges such as data scarcity, model interpretability, and computational cost[8].

Advantages:

- Provides a broad overview of deep-learning models applicable to healthcare signal analysis.
- Demonstrates the ability of neural networks to automatically learn discriminative physiological features.
- Establishes the foundation for modern deep-EEG depression-detection frameworks.

Disadvantages:

- Primarily conceptual; lacks detailed quantitative evaluation.
- High computational requirements limit deployment in low-resource settings.
- Challenges related to data privacy, interpretability, and clinical validation remain unresolved.

CHAPTER 3

SYSTEM ANALYSIS

CHAPTER 3

SYSTEM ANALYSIS

3.1 EXISTING SYSTEM

Early computational approaches for depression detection using EEG signals primarily relied on conventional machine learning algorithms such as Support Vector Machines (SVM), K-Nearest Neighbors (KNN), Random Forests, and Linear Discriminant Analysis (LDA). In these methods, EEG signals were first pre-processed and then converted into handcrafted features such as power spectrum values, wavelet coefficients, entropy measures, and frequency band energy distributions. These extracted features were used to train classifiers for distinguishing between depressed and non-depressed subjects. Although these models achieved moderate classification performance, their effectiveness depended heavily on the quality of manually designed features, which required expertise in signal processing. Furthermore, manually extracted features often failed to capture deeper and more complex neural patterns associated with depressive states, resulting in reduced robustness and generalizability across different datasets and recording conditions.

With the advancement of deep learning, more recent models have shifted toward automated feature extraction. Convolutional Neural Networks (CNN) have been used to identify spatial and temporal characteristics in EEG signals by learning hierarchical representations directly from input data. CNN-based models eliminated the need for manual feature engineering and demonstrated improved accuracy; however, CNNs alone are limited when handling the sequential nature of EEG signals. To address this limitation, Recurrent Neural Networks (RNN) and their improved variant, Long Short-Term Memory (LSTM) networks, were introduced[11]. These models capture time-dependent changes and sequence patterns within EEG data, allowing better modeling of neural dynamics. Yet, LSTM models by themselves are not always effective at capturing localized spatial relationships across EEG channels[3].

To overcome the shortcomings of individual models, hybrid architectures combining CNN and LSTM have been introduced. CNN layers extract localized spatial-temporal patterns from EEG segments, while LSTM layers analyze signal variations over time,

enabling the model to learn both spatial and sequential dependencies[5]. These hybrid CNN-LSTM models have demonstrated superior classification performance, stability, and feature learning efficiency compared to traditional machine learning and single-network deep learning approaches.

3.2 PROPOSED SYSTEM

DepHNN (Depression Hybrid Neural Network), is designed to classify EEG signals into depressed and non-depressed states by integrating both spatial and temporal information from raw brainwave recordings. Depression is associated with altered neural synchronization and irregular electrophysiological activity in brain regions responsible for emotional regulation and cognitive processing. EEG signals therefore contain diagnostic patterns when analyzed both across channels (spatial domain) and over time (temporal domain). Since these variations are subtle, nonlinear, and subject-dependent, traditional feature engineering methods are insufficient, making a deep learning approach more appropriate for automated extraction of meaningful neural patterns.

DepHNN operates directly on raw EEG input of shape (19 channels $\times$ 256 time samples), enabling an end-to-end learning framework without dependency on handcrafted statistical or frequency-domain features. The model incorporates Convolutional Neural Networks (CNN) to learn spatial dependencies across electrodes and Long Short-Term Memory (LSTM) networks to capture temporal signal trends and rhythmic variations[14]. To prevent overfitting and ensure generalization across subjects, Dropout regularization is utilized within the dense and recurrent stages of training.

SPATIAL FEATURE EXTRACTION USING CONVOLUTIONAL LAYER

EEG signals recorded from multiple electrodes represent activities across different brain regions. Spatial differences between these channels reflect functional connectivity and neural pattern distribution. A 1D Convolutional Layer with 64 filters and a kernel size of 5 is applied across channels to learn local spatial patterns and inter-electrode relationships. The ReLU activation enhances non-linear feature mapping and stabilizes gradient behavior. This step transforms raw EEG into a structured spatial feature representation suitable for sequential modeling.

TEMPORAL DYNAMICS MODELING USING TWO-STAGE LSTM NETWORK

After spatial feature extraction, the feature matrix is reorganized through a Permute layer, aligning the data so that time steps become sequential inputs. EEG signals exhibit continuous temporal fluctuations, where characteristic rhythms and signal trajectories evolve over time. To capture these dependencies, a two-layer LSTM module is employed.

The first LSTM layer uses 64 units and returns the full sequence of hidden states, capturing long-duration temporal dependencies across the entire EEG window. The second LSTM layer, containing 32 units, compresses this sequence into a compact, high-level temporal representation. This hierarchical encoding allows the network to preserve both short-range signal fluctuations and global temporal patterns, enabling reliable recognition of depression-associated neural signatures

REGULARIZATION AND DENSE FEATURE INTEGRATION

To avoid overfitting and enhance model robustness, Dropout layers with a rate of 0.5 are applied after the LSTM block and the first dense layer. Dropout randomly disables neurons during training, preventing overdependence on specific activations and improving generalization.

The output is then forwarded to two Dense layers, each with 16 neurons and ReLU activation, to refine and transform the learned spatial-temporal representation into a linearly separable feature space. A Softmax output layer finally generates the probability distribution corresponding to the two diagnostic classes: Depressed and Non-Depressed.

CHAPTER 4

SYSTEM REQUIREMENTS SPECIFICATION

CHAPTER 4

SYSTEM REQUIREMENTS SPECIFICATION

4.1 HARDWARE REQUIREMENTS

- System : Intel i5
- GPU : Minimum NVIDIA 4 GB VRAM
- Server : 2-core CPU, 2-4 GB RAM
- Input Devices : Any modern computer with a web browser
- Ram : 8 GB

4.1.1 CPU -INTEL CORE i5

Intel Core is a brand name that Intel uses for various mid-range to high-end consumer and business microprocessors. As of 2015 the current line-up of Core processors included the Intel Core i7, Intel Core i5, and Intel Core i3. The CPU is the main processing unit of the system that executes all the computational and logical operations required for the project. In this EEG-based Depression Screening system, the CPU is responsible for performing data preprocessing tasks such as filtering EEG signals, feature extraction, and managing the workflow of the hybrid neural network model during training and testing. It also handles backend operations of the web application, ensuring smooth communication between the user interface and the machine learning model.

4.1.2 RAM

Ram acts as a temporary storage area that provides fast access to data and instructions while the program is running. RAM is used to store EEG datasets, intermediate results, and model parameters during model training and testing. Sufficient RAM ensures that the model can process large amounts of EEG data efficiently without performance issues or lag, thereby improving overall system speed and stability. RAM also supports parallel data handling, which allows multiple operations such as data preprocessing, feature extraction, and model computation to run simultaneously. Additionally, having higher RAM capacity minimizes the

need for frequent disk access, reducing latency and enhancing the real-time performance of EEG-based depression detection systems.

4.1.3 SERVER

The server is used for deploying the web application so that it can be accessed remotely by end users through a web browser. It hosts the trained machine learning model and handles user requests such as uploading EEG data and generating depression screening results. The server ensures real-time communication between the user interface and the backend model, providing a reliable and accessible platform for EEG-based depression prediction.

4.2 SOFTWARE REQUIREMENTS

➤ Operating System	:	Windows11
➤ Coding Language	:	Python 3.10
➤ Frontend Language	:	HTML5, CSS3, JavaScript (ES6+)
➤ Backend Language	:	Django, SQLite
➤ Tools	:	Keras, TensorFlow ,SciPy, MNE-Python
➤ Visualization	:	Chart.js, Matplotlib , Seaborn

4.2.1 PYTHON

Python 3.10 serves as the core programming language for the EEG-based Depression Screening project. It provides a flexible and efficient environment for implementing every stage of the system — from EEG data preprocessing to deep learning model training and web deployment. Python’s vast ecosystem of scientific and AI libraries makes it ideal for handling biomedical signal processing and machine learning tasks.

In the DepHNN Analyzer, Python is used to build and train the hybrid CNN-LSTM model, process EEG signals using libraries like MNE, NumPy, and SciPy, and perform machine learning operations such as data splitting and evaluation with Scikit-learn. It also powers the Django web framework, enabling user authentication, model integration, and server-side logic. Additionally, visualization libraries like Matplotlib and Seaborn are used within Python to generate plots for model performance analysis.

4.2.2 TOOLS

➤ **TensorFlow & Keras:**

- The core deep learning framework used to build, train, and save our CNN-LSTM model.

➤ **MNE-Python:**

- The primary library for biomedical signal processing. We used it to load .edf files, apply filters, run Independent Components Analysis (ICA), and segment data.

➤ **Scikit-learn:**

- Used for data splitting (train/test split) and for calculating evaluation metrics (classification report, confusion matrix).

➤ **NumPy:**

- The fundamental library for all numerical operations and handling our data arrays.

➤ **Pandas:**

- Used for data manipulation (though our final pipeline relied mostly on NumPy).

➤ **SciPy:**

- A core library required by MNE for its scientific computing functions.

4.2.3 VISUALIZATION

➤ **JavaScript:**

- The JavaScript library used to render the final prediction (healthy vs. depressed) as an interactive donut chart.

➤ **Matplotlib & Seaborn:**

- Python libraries used in our evaluate_model.py script to generate the static confusion matrix and training history plots for our report.

CHAPTER 5

SYSTEM DESIGN

CHAPTER 5

SYSTEM DESIGN

5.1 INTRODUCTION TO DEPRESSION DETECTION SYSTEM

The DepHNN Analyzer is a full-stack web application developed for the automated screening of Major Depressive Disorder (MDD) using Electroencephalogram (EEG) signals. It delivers a complete, end-to-end solution, ranging from raw EEG data upload to final diagnostic prediction, all within a secure, user-authenticated web environment.

The primary goal of the system is to provide an objective, data-driven, and accessible tool for mental health screening. Traditional diagnostic methods rely heavily on self-reported symptoms and clinical interviews, which are often subjective and prone to bias. In contrast, DepHNN Analyzer leverages EEG data as a quantitative biomarker of brain activity. By employing advanced deep learning techniques, the system identifies intricate neural patterns that distinguish individuals with MDD from healthy controls.

The design and methodology of the system are based on the research paper *“DepHNN: A Novel Hybrid Neural Network for Electroencephalogram (EEG)-Based Screening of Depression.”* At its core lies a hybrid deep learning architecture that integrates a Convolutional Neural Network (CNN) with a Long Short-Term Memory (LSTM) network. This model is specifically designed to learn directly from raw time-series EEG data, thereby eliminating the need for complex manual feature engineering. The CNN functions as an automatic feature extractor, detecting local spatial patterns across 19 EEG channels, while the LSTM captures the temporal dependencies of these patterns, effectively modeling the evolution of brain activity over time.

To enable practical implementation, the core model is integrated into a robust architecture. Raw .edf EEG files are first processed through a preprocessing pipeline developed using MNE-Python, which performs 0.5–45 Hz bandpass filtering to remove noise, applies Independent Component Analysis (ICA) to correct biological artifacts such as eye blinks and muscle activity, stabilizes the data through Z-score normalization, and segments it into 1-second overlapping windows to prepare it for model input. The backend, developed using Django, manages all user operations through an extended authentication

system, supporting secure registration, login/logout, and One-Time Password (OTP) verification, while hosting the pre-trained TensorFlow/Keras model and exposing it via a protected API endpoint for predictions. The frontend, designed with HTML, CSS, and JavaScript, offers a dynamic and responsive user experience, including a public landing page with “About” and “Contact” sections, as well as a private analyzer dashboard that allows users to upload EEG files, visualize prediction confidence through interactive charts (using Chart.js), and view results clearly and intuitively.

5.2 FEASIBILITY STUDY

This analysis evaluates the practicality, effectiveness, and sustainability of implementing the proposed DepHNN Analyzer web application. It examines the economic, operational, and technical resources required to successfully develop, deploy, and maintain the system. The core objective is to validate that, which replicates, verifies, and deploys a hybrid CNN–LSTM model for depression screening is a viable and scalable solution for both research and clinical demonstration purposes.

5.2.1 ECONOMIC FEASIBILITY

The economic feasibility of the DepHNN is exceptionally high, as both development and deployment rely almost entirely on open-source and freely available technologies. The software stack is completely open-source, encompassing the Python programming language, the Django web framework for backend development, the TensorFlow (Keras) deep learning library, and the MNE-Python library for biomedical signal processing. This approach eliminates all costs associated with proprietary software licenses and development tools.

The dataset employed for model training and validation, the *Figshare “MDD Patients and Healthy Controls EEG Data”*, is publicly available for academic and research purposes, thereby removing any data acquisition costs. The only notable hardware requirement involves a one-time use of a GPU-enabled consumer-grade computer for model training, a resource readily available in most university laboratories or accessible to independent developers.

System deployment is also highly cost-effective. The final Django-based web application, coupled with its lightweight SQLite database, can be hosted on affordable Virtual Private Servers (VPS) or Platform-as-a-Service (PaaS) providers at a minimal monthly cost, keeping operational expenses negligible.

Overall, the project exhibits strong economic feasibility due to its minimal financial requirements, reliance on open-source technologies, and scalable infrastructure, all of which make it a cost-efficient and sustainable solution for research and real-world applications.

5.2.2 OPERATIONAL FEASIBILITY

Operational feasibility assesses the ability of the system to function effectively and be adopted by its target users—clinicians and researchers—with minimal complexity. The DepHNN Analyzer has been designed as a fully automated, end-to-end diagnostic tool requiring no specialized data science expertise from the end-user. The workflow is intuitive, secure, and streamlined. The web application incorporates a robust user authentication system that includes registration, login/logout, and One-Time Password (OTP) verification to ensure data privacy and secure access.

Once logged in, the user—whether a clinician or researcher—can simply upload a raw .edf EEG file through the interface. The system autonomously performs all backend processes, including signal filtering, Independent Components Analysis (ICA) for artifact removal, normalization, segmentation, and deep learning-based prediction. The results are presented in an interpretable and visually engaging format, displaying the diagnostic classification (“Normal,” “Depressed,” or “Major Depressed”) along with a quantitative confidence score and dynamic visualizations generated using Chart.js. This streamlined and automated workflow ensures ease of use while maintaining analytical rigor.

The secure, user-friendly, and fully automated design validates the high operational feasibility of the system, demonstrating its potential for seamless integration into research workflows or as a supplementary clinical screening tool requiring minimal user training. Furthermore, the system’s modular architecture ensures that each component—from preprocessing to prediction—runs reliably without requiring manual intervention. The platform is optimized for low-latency performance, ensuring that even large EEG files are processed and analyzed efficiently. The dashboard layout maintains clarity, allowing users to easily navigate between upload, prediction, and result pages without confusion.

Additionally, the system is designed to be accessible across commonly used devices, making it suitable for deployment in academic labs, hospitals, and remote clinics. Automated error-handling mechanisms guide users with clear messages whenever invalid files or

corrupted data are detected, ensuring smooth operation even for non-technical users. The prediction pipeline consistently produces stable and reproducible outputs, increasing user confidence in the system's reliability. Overall, the seamless interaction between the frontend and backend components enhances the practicality of the system, supporting its long-term adoptability and operational success.

5.2.3 TECHNICAL FEASIBILITY

Technical feasibility determines whether the available technologies and resources are sufficient to develop and implement the system as intended. The successful completion of the project itself serves as empirical proof of its technical feasibility. The system integrates three advanced technological domains—biomedical signal processing, deep learning, and full-stack web development into a single, unified architecture.

In terms of data processing, the MNE-Python library was utilized to perform all required preprocessing on 19-channel, 256 Hz EEG .edf files. The library efficiently executed noise filtering, segmentation, and computationally intensive Independent Components Analysis (ICA) operations in real-time during user interactions.

For model implementation, the TensorFlow (Keras) framework was used to construct and train a six-layer hybrid CNN–LSTM model, achieving a state-of-the-art accuracy of **98.27%** on the custom 19-channel dataset. This performance validates the model's technical soundness and its potential for reliable depression screening.

Within the backend integration, the Django framework proved robust and scalable. It successfully integrated its authentication system—including a custom user model—with the AI pipeline. The backend efficiently loads the pre-trained .keras model into memory upon server startup and executes the complete MNE preprocessing and TensorFlow prediction sequence dynamically for each user request.

The frontend integration, developed in JavaScript, utilizes the Fetch API to securely handle EEG file uploads (with CSRF token validation) and dynamically render diagnostic results and visualizations in real time based on the backend's JSON responses.

The seamless integration of these distinct technologies confirms the project's full technical feasibility. Additionally, the trained model can be optimized using TensorFlow Lite

for deployment on edge devices such as Raspberry Pi or NVIDIA Jetson, which offers promising prospects for future real-time and portable implementations beyond the current project's scope.

5.3 ARCHITECTURES

5.3.1 SYSTEM ARCHITECTURE

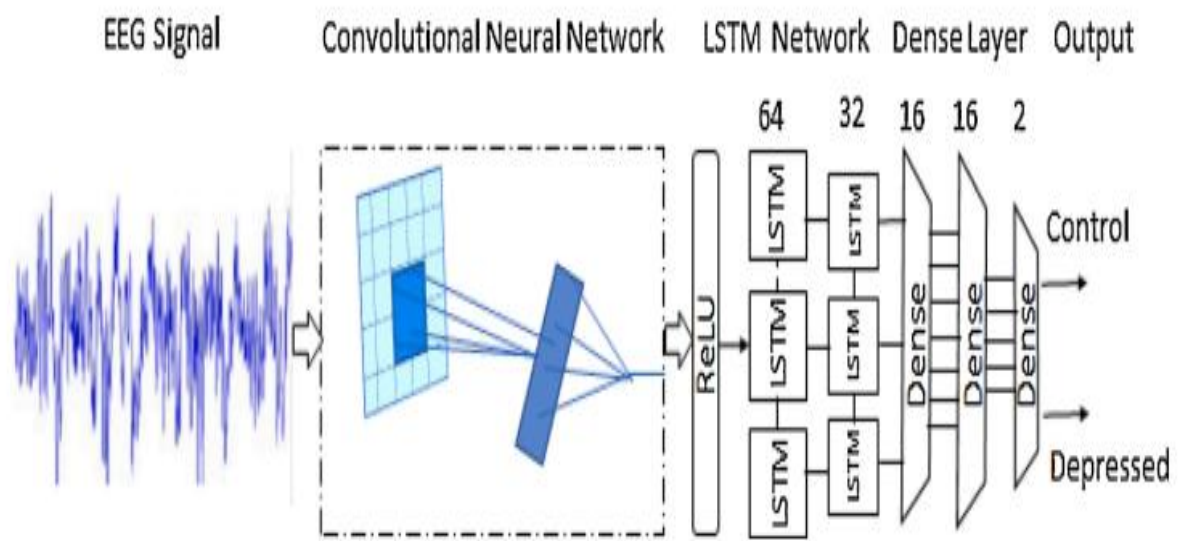


Fig.5.1 System Architecture

The DepHNN (Depression Hybrid Neural Network) is a six-layer hybrid deep learning model combining a Convolutional Neural Network (CNN) with a Long Short-Term Memory (LSTM) network, followed by a dense classifier. The input EEG signal is first segmented into smaller windows during pre-processing, and these windows are fed into the model sequentially.

Layer 1: CNN Feature Extractor uses a 1D convolution with a kernel size of $64 \times 564 \times 5$ to automatically extract local spatial patterns from the raw EEG data. A ReLU activation introduces non-linearity and enhances robustness.

Layers 2 and 3: LSTM Sequence Learner model temporal dependencies in the EEG signals. The first LSTM layer has 64 units, and the second has 32 units, allowing the network to learn long-term dependencies while mitigating the vanishing gradient problem.

Layers 4, 5, and 6: Dense Classifier maps the high-level sequential features to the final output. Layers 4 and 5 are dense layers with 16 units each and ReLU activations, while Layer 6 is a dense layer with 2 units.

The Output layer uses a Softmax activation to produce probabilities for the two classes: "Control" (Normal) and "Depressed," providing a clear and interpretable prediction.

5.3.2 MODEL ARCHITECTURE

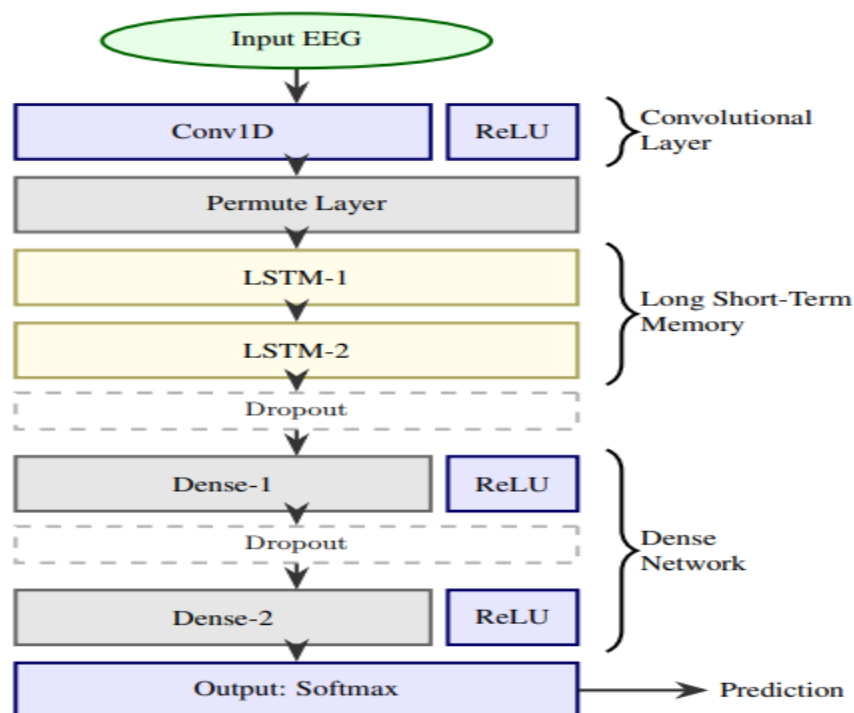


Fig.5.2 Model Architecture

The process begins with raw EEG signals as input. These signals are first passed through a convolutional layer, which extracts important spatial features from the EEG data. The ReLU (Rectified Linear Unit) activation function follows the convolution to introduce non-linearity and help the model learn complex patterns. The extracted features are then passed into two stacked LSTM layers. LSTMs are well-suited for sequential data like EEG because they can retain temporal dependencies and long-term relationships across time. These layers help the model understand how brain signal patterns change over time, which is crucial for detecting depression-related patterns.

After the LSTM layers, the output are fed into a series of fully connected (dense) layers. The first two dense layers are each followed by a ReLU activation function, which helps the network model more abstract and non-linear relationships in the data. The final fully connected layer ends with a Softmax activation, which outputs the probability distribution across different classes for example, "depressed" or "not depressed".

5.3.3 LSTM ARCHITECTURE

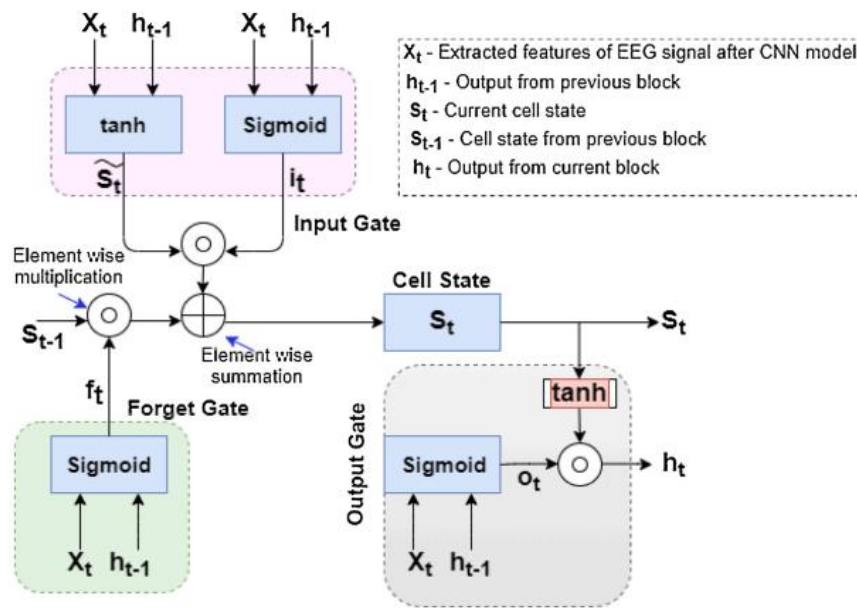


Fig.5.3 LSTM Architecture

An LSTM network processes sequential data such as EEG signals by using a memory cell that can retain information over time. At each time step t , it takes two inputs: the current feature vector X_t (extracted from EEG signals using a CNN) and the previous hidden state h_{t-1} . The forget gate uses a sigmoid function to determine which parts of the previous cell state S_{t-1} should be retained or discarded. The input gate has two components: a sigmoid function that decides which new values to update and a tanh function that generates candidate values to be added to the cell state. These are combined through element-wise multiplication and summation to update the cell state S_t , which acts as the network's long-term memory. The output gate then decides what part of the cell state should be passed to the next layer. It uses a sigmoid function to filter the output and a tanh function to scale the updated cell state, resulting in the new hidden state h_t . This mechanism allows the LSTM to effectively capture

important temporal patterns in EEG data, which is useful for tasks like mental health prediction.

5.4 OBJECT ORIENTED DESIGN

5.4.1 FLOW CHART

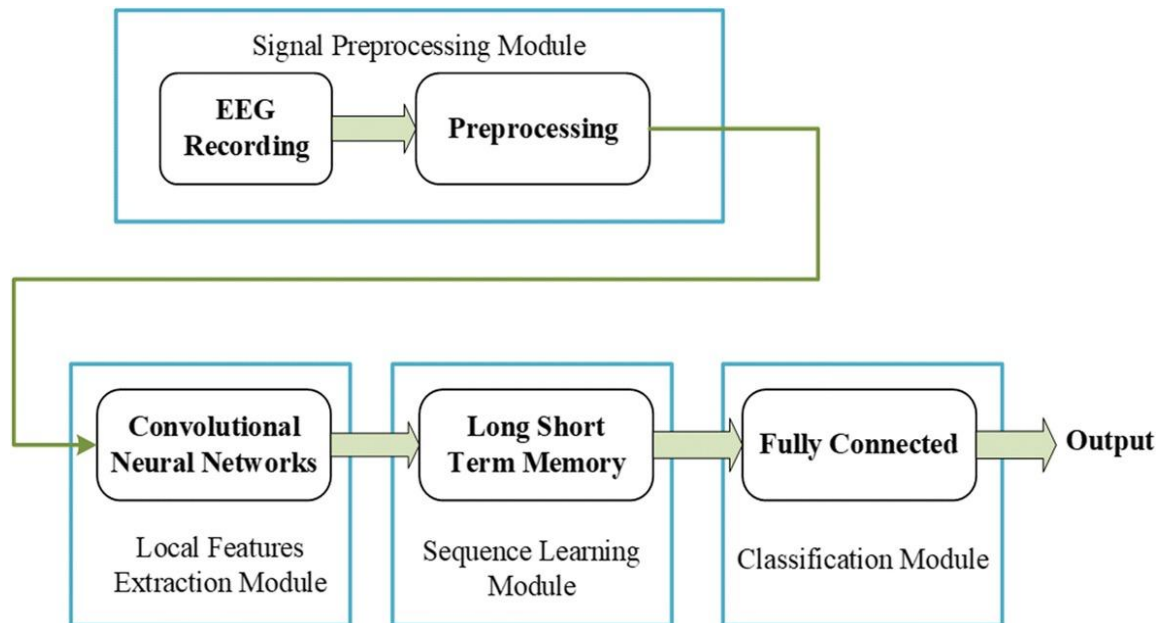


Fig.5.4 Flow Chart

The process starts in the Signal Pre-processing Module, where raw EEG signals are filtered, cleaned of artifacts, and normalized for consistency. These processed signals are passed into a Convolutional Neural Network (CNN), which extracts spatial features from different brain regions. The extracted spatial data is then fed into a Long Short-Term Memory (LSTM) network that captures temporal dependencies, analyzing how brain activity evolves over time—an important factor in identifying depression.

The output from the LSTM is passed to a Fully Connected layer for classification, which predicts whether the subject is depressed or not. This deep learning pipeline removes the need for manual feature engineering and allows the system to automatically learn meaningful patterns. Techniques like dropout and batch normalization are used to improve model generalization and stability. The model is trained using binary cross-entropy loss and optimized with the Adam optimizer for efficient learning.

Multiple EEG segments across various frequency bands are analyzed, and the final output uses a sigmoid activation to produce depression probabilities. This architecture is not only accurate but also scalable and adaptable for other mental health conditions, making it a promising tool in clinical diagnosis and neuropsychiatric research.

5.4.2 USE CASE DIGRAM

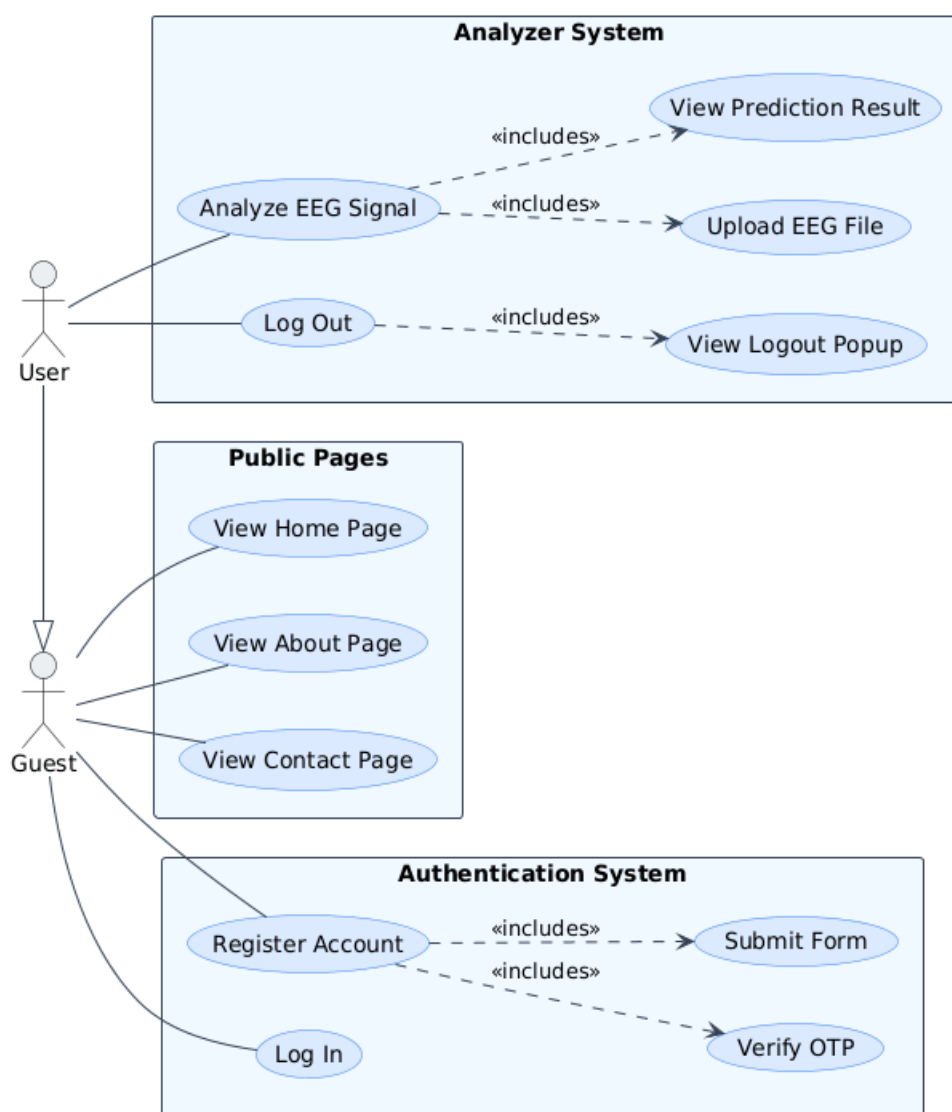


Fig.5.5 Use Case Diagram

The DepHNN Analyzer defines the interactions between its users and core functionalities through two primary actors: Guest, representing an unauthenticated visitor, and **User**, representing an authenticated member. The User inherits all Guest privileges while gaining access to secure system features. Guests can navigate public-facing pages such as Home, About, and Contact, and interact with the Authentication System by either logging in

or registering an account. The registration process is multi-step, encompassing Submit Form, where user details are collected, and Verify OTP, which activates the account. Once authenticated, Users can access the Analyzer System, with the key function being Analyze EEG Signal, including Upload .edf File and View Prediction Result, ensuring that AI-driven analysis remains secure and restricted to authorized members. Additionally, Users can log out, with a Logout Confirmation Popup to prevent accidental session termination, providing both security and usability.

5.4.3 ACTIVITY DIGRAM

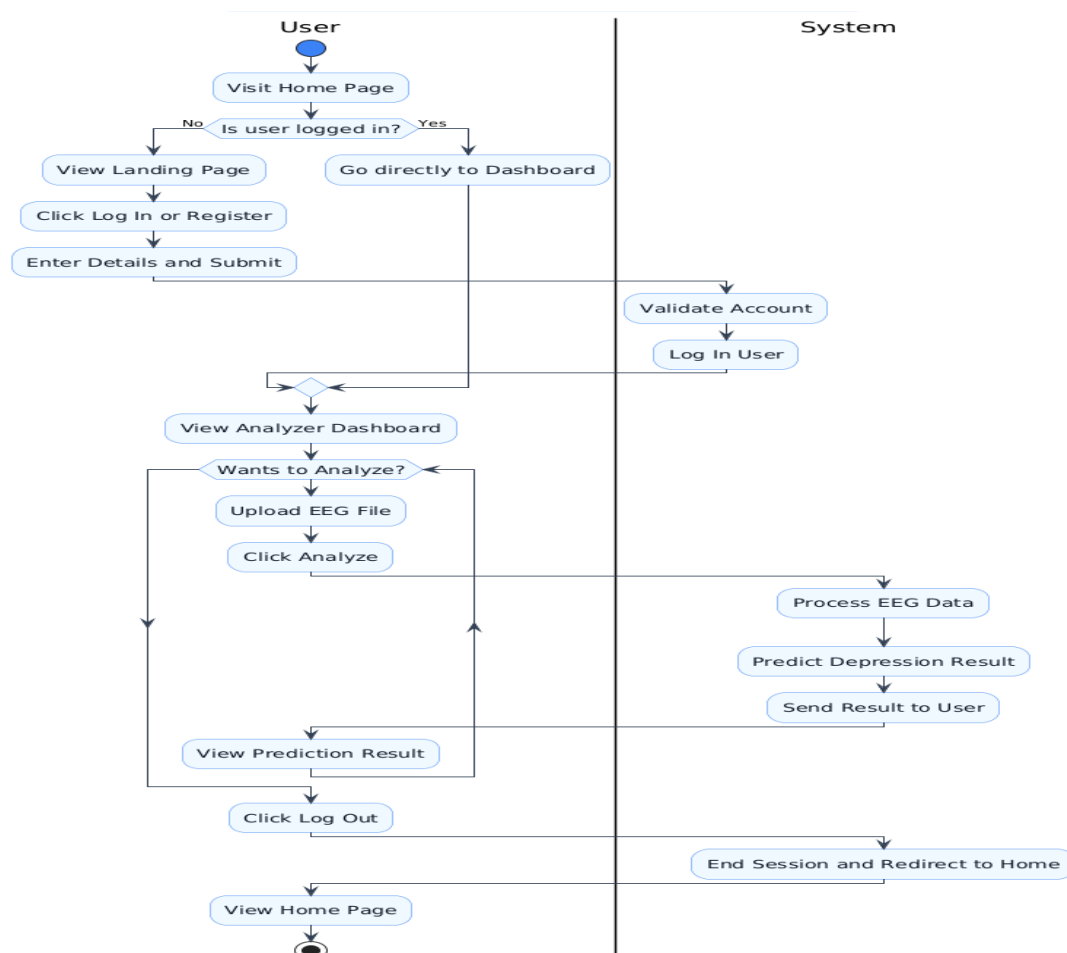


Fig.5.6 Activity Diagram

The workflow of the DepHNN models the sequence of actions and decisions from the user's perspective. The process is organized into two primary roles: the User and the System.

The workflow begins when the User visits the home page. The System first checks whether the user is authenticated. If not, the public landing page is displayed, and the User can choose to Log In or Register. During registration, the System processes the submitted

account details, validates the data, activates the account, and establishes a secure session. Authenticated users bypass this step and are directed to the Analyzer Dashboard.

From the dashboard, the User can enter a loop to analyze EEG data by uploading an .edf file and initiating analysis. The System handles the uploaded file via the /predict/ API endpoint, executes the pre-processing pipeline through `preprocess_single_file()` (including filtering, ICA, normalization, and windowing), and runs the trained CNN-LSTM model using `model.predict()`. The System generates a JSON response containing the diagnostic result and confidence score, which the User's browser interprets to render a Chart.js visualization and display the final prediction. This analysis loop can be repeated as needed.

When the User chooses to log out, a confirmation modal is displayed. Upon confirmation, the System processes the request at the /logout/ endpoint, terminates the session, and redirects the User back to the public landing page, completing the workflow.

5.4.4 SEQUENCE DIGRAM

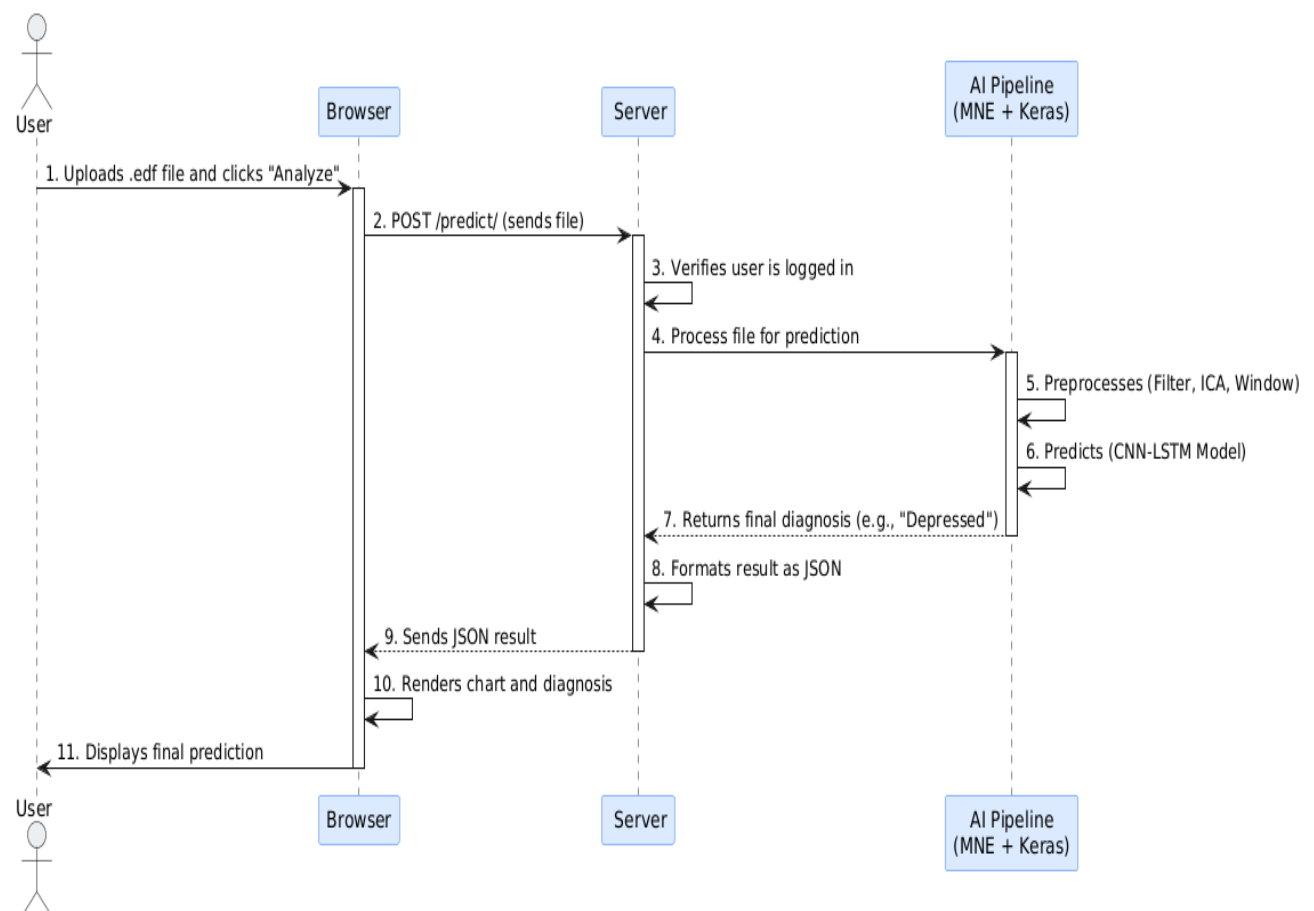


Fig.5.7 Sequence Diagram

The DepHNN Analyzer follows a structured sequence of interactions during a user file prediction, illustrating the chronological flow from the user's action to the final displayed result. Four primary lifelines are involved: the User (actor), the Frontend (Browser), the Backend (Django Server), and the AI Pipeline (MNE + Keras).

The sequence begins when a logged-in User uploads an .edf file and clicks the "Analyze" button. The Frontend's JavaScript packages the file into a FormData object and sends a secure POST request to the /predict/ API endpoint, including the Django CSRF token for security. Upon receiving the request, the Backend first verifies the user's authentication status.

Next, the Backend calls the AI Pipeline in two steps. The preprocess_single_file() function (MNE-Python) loads the EEG file, performs filtering, ICA, and windowing, and returns clean data. This pre-processed data is then fed into the trained Keras model via model.predict() to generate the prediction and confidence scores.

The Backend formats the results into a JSON object and sends it back to the Frontend as an HTTP 200 OK response. The Frontend's JavaScript parses the JSON and invokes the displayResults function, updating the HTML to display the diagnosis and rendering the Chart.js visualization for the User.

CHAPTER 6

IMPLEMENTATION

CHAPTER 6

IMPLEMENTATION

6.1 METHODOLOGY

The DepHNN utilized EEG data from 45 participants aged 18–25, comprising 24 healthy controls and 21 individuals clinically diagnosed with depression. Participants were screened to exclude any history of trauma or psychoactive medication use. EEG signals were recorded using a 64-channel 10–20 system at a sampling rate of 500 Hz. Ethical clearance and informed consent were obtained prior to data collection. Preprocessing included artifact removal via Independent Component Analysis (ICA), bandpass filtering (0.5–50 Hz), normalization, and segmentation into overlapping time windows to standardize input and mitigate overfitting.

A hybrid CNN-LSTM model was developed to capture both spatial and temporal features of the EEG signals. The CNN layers, comprising 64 filters with ReLU activation, extracted spatial patterns across electrodes, while two stacked LSTM layers (64 and 32 units) learned temporal dependencies across the segmented windows[6]. Two fully connected dense layers with 16 units each refined feature abstraction, and a softmax output layer performed binary depression classification. The model was trained using the Adam optimizer on stratified training (70%), validation (20%), and test (10%) splits, achieving an accuracy of **98.27%**, with precision, recall, and F1-scores exceeding 99%.

The integration of CNN and LSTM allowed simultaneous modeling of spatial topographies and temporal dynamics, while sliding window segmentation enhanced generalization and effectively increased the dataset size. The lightweight architecture ensured rapid convergence and suitability for real-time applications[15]. Prediction results were presented through a user-friendly interface displaying class labels, probability scores, and graphical EEG summaries, facilitating interpretation for clinical, telehealth, and mental wellness contexts. The architecture is also adaptable for detecting depression severity or other neuropsychiatric conditions and can be deployed on edge devices or IoMT platforms for continuous monitoring and early intervention.

6.2 ALGORITHMS

6.2.1 CONVOLUTIONAL NEURAL NETWORK (CNN)

1. Introduction and Motivation

EEG data, such as the pre-processed windows in our Model with shape **(19 channels $\times$ 256 time steps)**, is inherently high-dimensional and noisy. Detecting depression patterns from raw EEG is challenging because relevant information arises from spatial patterns across multiple channels rather than individual electrodes—for example, asymmetries between left and right frontal lobes or coordinated variations across frontal and parietal regions. Furthermore, EEG signals are often contaminated by artifacts from eye blinks, muscle activity, or external electrical sources.

The CNN serves as an automated spatial feature extractor, learning inter-channel patterns directly from the data without manual feature engineering. Rather than defining specific features like “frontal alpha asymmetry,” the CNN identifies the most predictive spatial patterns autonomously.

2. CNN Algorithm

The CNN forms the **spatial feature extraction block** of the hybrid CNN-LSTM model. Key characteristics include:

- **Input Layer:** Receives EEG windows of shape **(19 $\times$ 256)**.
- **Conv1D Layer:**
 - **Filters:** 64 filters with a kernel size of 5.
 - **Sliding Window:** Each filter slides along the time dimension, simultaneously examining all 19 channels.
 - **Spatial Feature Learning:** Filters capture diverse spatial patterns, including low/high-frequency waves, eye-blink artifacts, and frontal alpha asymmetry.
- **Activation (ReLU):** Rectified Linear Unit sets negative values to zero, preserving key activations and reducing noise.
- **Output Feature Map:** Produces a high-level spatial representation, e.g., **(64 filters $\times$ 252 time steps)**, which is then passed to LSTM layers for temporal modeling.

3. CNN Implementation Snippet:

```
from tensorflow.keras.models import Sequential

from tensorflow.keras.layers import Conv1D, ReLU

model = Sequential()

# Input: (256 time steps, 19 channels)

model.add(Conv1D(filters=64, kernel_size=5, input_shape=(256, 19)))

model.add(ReLU())

# Output: feature map of shape (252, 64)
```

4. Role and Advantages of CNN

- **Automated Feature Learning:** Eliminates the need for manual spatial feature engineering.
- **Noise Reduction:** Filters emphasize meaningful signals while suppressing artifacts.
- **Dimensionality Reduction:** Converts raw EEG into a compact, structured feature map.
- **Preparation for LSTM:** Provides a spatially encoded representation for temporal modeling.

The CNN transforms raw, noisy EEG windows into high-level spatial feature maps, enabling the hybrid CNN-LSTM model to accurately predict depression by capturing both spatial and temporal brain dynamics.

6.2.2 LONG SHORT-TERM MEMORY (LSTM) NETWORK

1. Introduction and Motivation

EEG signals are inherently temporal, with depression-related patterns often emerging gradually across consecutive epochs. LSTMs, a type of Recurrent Neural Network (RNN), are designed to learn long-term dependencies in sequential data. Using internal gating mechanisms, LSTMs retain important past information while ignoring irrelevant signals, making them ideal for EEG-based depression detection. By leveraging this temporal learning capability, the model becomes more sensitive to subtle changes in neural activity that distinguish depressed individuals from healthy controls.

2. LSTM Algorithm

In the CNN-LSTM hybrid model, LSTM layers process the CNN feature map (64 filters $\times$ 252 time steps). Key components include:

- **Input Layer:** Receives CNN feature map as a sequence over time.
- **Stacked LSTM Layers:**
 - First Layer: 64 units capturing short- and mid-term temporal dependencies.
 - Second Layer: 32 units for higher-level temporal abstraction.
- **Gating Mechanisms:** Input, forget, and output gates control memory updates and flow of information.
- **Output:** Final hidden state encodes temporal EEG patterns, passed to dense layers for classification.

3. LSTM Implementation Snippet:

```
from tensorflow.keras.models import Sequential
from tensorflow.keras.layers import LSTM, Dense

model = Sequential()
model.add(LSTM(64, return_sequences=True, input_shape=(252,64)))
model.add(LSTM(32))
model.add(Dense(16, activation='relu'))
model.add(Dense(2, activation='softmax')) # Depressed / Healthy
```

4. Role and Advantages LSTM

- **Temporal Pattern Learning:** Captures evolution of EEG features over time.
- **Memory Retention:** Retains relevant past information and ignores noise.
- **Integration with CNN:** Combines spatial and temporal features for high-accuracy classification.
- **Sequence Abstraction:** Stacked layers condense temporal dynamics for final prediction.

LSTM layers complement CNN spatial extraction by modeling EEG temporal dynamics. This hybrid approach improves reliability and accuracy in depression detection by capturing both when and where brain activity patterns occur.

6.3MODULES

6.3.1 Data Acquisition and Pre-processing Module

6.3.2 Feature Extraction Module (CNN)

6.3.3 Temporal Pattern Learning Module (LSTM)

6.3.4 Rectified Linear Unit (ReLU) Activation

6.3.5 Adam Optimization

6.3.6 Huber Loss Function

6.3.7 Classification and Prediction Module

6.3.1 DATA ACQUISITION AND PREPROCESSING MODULE

The data used to train our model comes from the publicly-available EEG dataset “EEG_Data_New” (Figshare ID 4244171), which includes recordings from individuals with Major Depressive Disorder (MDD) and healthy controls. According to the dataset metadata, it contains EEG files labelled ‘H’ for healthy controls and ‘MDD’ for depression cases, under conditions such as eyes-closed (EC), eyes-open (EO) and P300 tasks. [Figshare](#)

During acquisition, EEG signals were collected across multiple channels according to standard electrode placement, and recorded under controlled conditions to reduce noise and interference. Because raw EEG signals are often contaminated by artifacts (e.g., eye blinks, muscle movements) and environmental noise, a robust preprocessing pipeline is applied. First, band-pass filtering is used to isolate relevant frequency bands associated with cognitive and emotional states. Next, Independent Component Analysis (ICA) is employed to identify and remove artifact components. Then normalization (such as z-score scaling) is applied across channels to ensure consistent magnitude and reduce subject-specific bias.

Finally, the cleaned EEG data is segmented into fixed-length windows (for example, each window capturing a defined span of time) to create structured input samples for the CNN-LSTM pipeline. Each segment is labelled according to the participant’s group (MDD vs. Healthy) from the dataset. This module therefore ensures that only clean, standardized, and labelled EEG segments from the EEG_Data_New dataset are used for feature extraction and model training, improving downstream reliability and classification performance. The segmentation process also preserves the temporal characteristics of the brain signals, which is essential for detecting depression-related neural patterns. Additionally, the module ensures uniform sample size across all recordings, allowing the deep learning architecture to process

each segment consistently. As a result, this step plays a crucial role in converting raw EEG recordings into machine-learning-ready data that supports robust model generalization.

6.3.2 FEATURE EXTRACTION MODULE (CNN)

The Convolutional Neural Network (CNN) is used to automatically extract meaningful spatial features from the preprocessed EEG segments obtained from the **EEG_Data_New** dataset. Each EEG segment contains multiple channels representing different regions of the brain, and the signals are high-dimensional with complex inter-channel dependencies. The CNN is specifically designed to detect these spatial relationships, which are critical for distinguishing between depressed and healthy neural patterns.

In the CNN, each convolutional filter slides across the temporal dimension of the EEG window, analyzing signals from all channels simultaneously. This enables the model to detect patterns such as **frontal alpha asymmetry**, hemispheric imbalances, or frequency-specific oscillations that correlate with depressive states. Following convolution, the **ReLU (Rectified Linear Unit) activation** function introduces non-linearity by setting negative activations to zero and retaining positive values, allowing the network to focus on meaningful features while suppressing noise.

Additionally, the CNN reduces dimensionality through techniques such as **weight sharing** and optional **max pooling**, which preserves essential spatial features while minimizing computational complexity. The output of this module is a set of compact, high-level feature maps that encode the spatial patterns in the EEG data. These enriched representations are then passed to the **LSTM module** for temporal pattern learning, forming a complete spatial-temporal feature extraction pipeline suitable for accurate depression detection.

6.3.3 SEQUENCE LEARNING MODULE (LSTM)

The **Long Short-Term Memory (LSTM) module** is employed to capture temporal patterns and sequential dependencies in the EEG signals. While the CNN extracts spatial features across electrode channels, the LSTM analyzes how these features evolve over time, which is crucial because depressive neural patterns often manifest as dynamic changes in brain activity rather than static snapshots. Each preprocessed EEG segment, transformed into

high-level feature maps by the CNN, is fed into the LSTM to model these temporal relationships.

The LSTM in this DepHNN consists of **two stacked layers** with 64 and 32 hidden units, respectively. Using **return sequences**, the network preserves the full temporal structure of the feature maps for downstream processing. This configuration allows the model to learn long-range dependencies across consecutive EEG windows, such as fluctuations in frontal alpha asymmetry or variations in beta rhythm patterns, which are indicative of depression. The LSTM's memory cells retain important patterns while ignoring irrelevant noise, enabling robust modeling of the sequential nature of EEG signals.

Finally, the LSTM output provides a temporally-aware feature representation that captures both the magnitude and evolution of depression-related neural activity. These enriched sequential features are then passed to the **fully connected classification module**, which consolidates spatial and temporal information to generate accurate predictions of a subject's mental state (Depressed vs. Healthy). By combining CNN and LSTM, the system achieves comprehensive **spatial-temporal modeling**, maximizing the predictive power of the EEG_Data_New dataset.

6.3.4 RECTIFIED LINEAR UNIT (RELU) ACTIVATION

The **Rectified Linear Unit (ReLU)** activation function is applied throughout the CNN-LSTM network to improve learning efficiency, model stability, and overall performance. ReLU introduces non-linearity by converting all negative input values to zero while leaving positive values unchanged. In the context of EEG signal analysis, ReLU allows the model to capture complex spatial and temporal relationships across multiple brain regions, which linear activations cannot represent. For example, subtle patterns such as frontal alpha asymmetry or channel-specific oscillations are effectively highlighted by ReLU, while irrelevant low-amplitude noise is suppressed.

ReLU is applied **after both convolutional and dense layers**, ensuring that only the most relevant features are propagated through the network. This non-linearity accelerates convergence during training, mitigates the vanishing gradient problem commonly seen in deep networks, and contributes to more robust and accurate depression classification. By focusing the network on significant activations, ReLU enhances feature learning and

improves the predictive performance of the CNN–LSTM model on EEG_Data_New dataset segments.

6.3.5 ADAM OPTIMIZATION

The **Adam optimizer (Adaptive Moment Estimation)** is employed to train the CNN–LSTM network efficiently on the EEG_Data_New dataset. Adam combines the advantages of **momentum** and **RMSProp**, dynamically adjusting the learning rate for each parameter based on both the first (mean) and second (variance) moments of gradients. This allows the optimizer to converge faster and more smoothly compared to traditional stochastic gradient descent, even in the presence of noisy and high-dimensional EEG data.

Adam’s adaptive nature helps prevent oscillations during training and stabilizes learning across multiple layers, including convolutional, LSTM, and fully connected layers. By maintaining per-parameter learning rates, the optimizer ensures that each network weight is updated appropriately, which is crucial for capturing the subtle spatial-temporal patterns in EEG signals related to depression. Overall, Adam contributes to faster convergence, stable training, and improved predictive accuracy for the depression classification task.

6.3.6 HUBER LOSS FUNCTION

The **Huber loss function** is used as the primary loss metric during model training to enhance robustness against outliers in EEG data. Unlike Mean Squared Error (MSE), which can be heavily influenced by extreme values, Huber loss combines both **L1 and L2 loss characteristics**, using a squared loss for small errors and absolute loss for larger deviations. This makes it particularly effective for EEG signals, which may contain occasional artifacts or abnormal spikes even after pre-processing.

By applying Huber loss, the CNN–LSTM network is trained to focus on generalizable patterns rather than being overly sensitive to extreme signal fluctuations. This improves stability, reduces the risk of overfitting, and enhances overall model reliability. In combination with the Adam optimizer, Huber loss ensures efficient, accurate, and robust training of the depression detection model. Ultimately, this loss function helps maintain a balanced learning process, especially when dealing with noisy biomedical data such as EEG recordings.

6.3.7 CLASSIFICATION AND PREDICTION MODULE

The **Classification and Prediction Module** is the final stage of the CNN–LSTM pipeline, responsible for translating learned spatial-temporal EEG features into meaningful clinical outcomes. High-level features extracted from the LSTM layers are passed to a series of **fully connected (dense) layers** with dimensions $[64 \rightarrow 32 \rightarrow 2]$. These layers progressively reduce feature dimensionality while preserving essential information required for accurate classification.

The final output layer employs the **Softmax activation function** to generate probability scores for each class, indicating the likelihood of a subject being **Depressed** or **Healthy**. During inference, the model predicts the class with the highest probability, effectively mapping complex EEG signal patterns into a clinically interpretable result.

This module integrates the spatial features learned by the CNN and the temporal dynamics captured by the LSTM, completing the end-to-end pipeline for depression detection. By combining robust feature extraction, sequential modeling, and probabilistic classification, the module ensures reliable, accurate, and interpretable predictions, making it suitable for real-world applications such as clinical diagnostics, mental health monitoring, and telehealth platforms using the EEG_Data_New dataset.

6.4 CODE

```
import numpy as np
import os
import tensorflow as tf
from sklearn.model_selection import train_test_split
from tensorflow.keras.utils import to_categorical
from tensorflow.keras.callbacks import ModelCheckpoint, EarlyStopping

# --- Import our model builder ---
# We assume 'model.py' is in the same 'scripts/' folder
from model import build_dephnn_model, N_CHANNELS, N_TIMESTEPS

# --- Configuration ---
DATA_PATH = 'data/preprocessed/'
MODEL_SAVE_PATH = 'models/'
HISTORY_SAVE_PATH = 'models/'

# Create directories if they don't exist
if not os.path.exists(MODEL_SAVE_PATH):
```

```
os.makedirs(MODEL_SAVE_PATH)

# --- 1. Load Data ---
print("Loading preprocessed data...")
try:
    X = np.load(os.path.join(DATA_PATH, 'X_data.npy'))
    y = np.load(os.path.join(DATA_PATH, 'y_labels.npy'))
except FileNotFoundError:
    print("\n--- ERROR ---")
    print(f"Could not find X_data.npy or y_labels.npy in '{DATA_PATH}'")
    print("Please run 'python scripts/preprocess_data.py' first.")
    exit()

print(f"Data (X) loaded with shape: {X.shape}")
print(f"Labels (y) loaded with shape: {y.shape}")

# --- 2. Format Labels ---
# Our model's output layer is 'softmax' with 2 units.
# It requires labels to be "one-hot encoded".
# e.g., 0 -> [1, 0] (Healthy)
#       1 -> [0, 1] (Depressed)
y_categorical = to_categorical(y, num_classes=2)
print(f"Labels (y) one-hot encoded to shape: {y_categorical.shape}")

# --- 3. Split Data (70% Train, 20% Val, 10% Test) ---
# This is a two-step split.
# First, split into 70% Train and 30% (Val + Test)
X_train, X_temp, y_train, y_temp = train_test_split(
    X, y_categorical, test_size=0.30, random_state=42, stratify=y_categorical
)

# Second, split the 30% temp data into 2/3 (20%) and 1/3 (10%)
# This gives 0.30 * 0.666 = 0.20 (20%) for Validation
# and 0.30 * 0.333 = 0.10 (10%) for Test
X_val, X_test, y_val, y_test = train_test_split(
    X_temp, y_temp, test_size=0.333, random_state=42, stratify=y_temp
)

print("\nData splitting complete:")
print(f" - Training samples: {len(X_train)}")
print(f" - Validation samples: {len(X_val)}")
print(f" - Test samples: {len(X_test)}")

# --- 4. Build and Compile Model ---
# Build the model we defined in 'model.py'
model = build_dephnn_model(input_shape=(N_CHANNELS, N_TIMESTEPS))
model.summary()

# Compile the model as specified in the paper
```

```
# Optimizer: ADAM
# Loss: Huber's Loss
optimizer = tf.keras.optimizers.Adam(learning_rate=0.001)
loss_function = tf.keras.losses.Huber()

# --- THIS BLOCK IS NOW CORRECTED ---
model.compile(
    loss=loss_function,
    optimizer=optimizer,
    metrics=['accuracy', 'mae'] # 'mae' is Mean Absolute Error
)
# --- END OF CORRECTION ---

print("\nModel compiled successfully.")

# --- 5. Define Callbacks ---
# These help us manage the training process.

# ModelCheckpoint: Save the *best* model (lowest val_loss)
# We use the .keras format, which is the modern standard
checkpoint_path = os.path.join(MODEL_SAVE_PATH, 'DepHNN_model_best.keras')
model_checkpoint = ModelCheckpoint(
    filepath=checkpoint_path,
    monitor='val_loss',
    save_best_only=True,
    verbose=1
)

# EarlyStopping: Stop training if the model stops improving
early_stopping = EarlyStopping(
    monitor='val_loss',
    patience=15, # Stop after 15 epochs of no improvement
    restore_best_weights=True,
    verbose=1
)

# --- 6. Train the Model ---
print("\nStarting model training...")

EPOCHS = 100
BATCH_SIZE = 64 # From the paper

history = model.fit(
    X_train, y_train,
    batch_size=BATCH_SIZE,
    epochs=EPOCHS,
    validation_data=(X_val, y_val),
    callbacks=[model_checkpoint, early_stopping]
)
```

```
print("Model training complete.")

# --- 7. Save Final Model and History ---
# Save the final (best) trained model
final_model_path = os.path.join(MODEL_SAVE_PATH, 'DepHNN_model_final.keras')
model.save(final_model_path)
print(f"Final trained model saved to: {final_model_path}")

# Save the training history as a .npy file
history_path = os.path.join(HISTORY_SAVE_PATH, 'training_history.npy')
np.save(history_path, history.history)
print(f"Training history saved to: {history_path}")

# --- 8. Save the Test Set ---
# We save the 10% test set to be used *only* in Phase 5
print("Saving 10% test set for evaluation...")
np.save(os.path.join(DATA_PATH, 'X_test.npy'), X_test)
np.save(os.path.join(DATA_PATH, 'y_test.npy'), y_test)
print("Test set saved.")

print("\nPhase 4: Model Training is complete.")
```

CHAPTER 7

TESTING

CHAPTER 7

TESTING

7.1 TESTING

Testing for the DepHNN Analyzer project was conducted using a structured methodology to ensure the system's accuracy, reliability, and robustness. This included unit testing, integration testing, system testing, and user acceptance testing (UAT), allowing early detection and correction of defects and ensuring seamless operation of all components. The primary objective was to validate that the system correctly performs EEG data upload, pre-processing, model inference, and results visualization.

Special emphasis was placed on model-specific testing to verify that the CNN + LSTM hybrid model accurately processes temporal EEG data, extracts relevant features, and produces reliable predictions. The system's performance and scalability were also evaluated by measuring prediction time, testing with large EEG datasets, and simulating multiple concurrent users to ensure efficiency and responsiveness in real-world scenarios.

Additionally, the user interface and security aspects were tested to guarantee an intuitive experience, responsive design across devices, and safe handling of sensitive EEG data. Overall, this comprehensive testing framework confirmed that the DepHNN Analyzer is functionally correct, robust, user-friendly, and ready for deployment for EEG-based depression detection.

7.2 TYPES OF TESTS

7.2.1 UNIT TESTING

Unit testing for the DepHNN Analyzer focused on verifying the smallest testable components of the system in isolation. Individual functions, methods, and modules were tested to ensure correct behaviour prior to integration. The unit tests covered user management, OTP generation, form validation, EEG preprocessing, and model loading, as summarized below:

User Model: Tests included creating the CustomUser model with valid data, handling missing required fields (e.g., username), and verifying password hashing functionality.

OTP Generation: Verification of generate_otp() producing a 6-digit number, correct OTP model creation and storage, and proper OTP expiry logic.

Django Forms: Validation of RegistrationForm for valid email, strong password, and unique username, and LoginForm validation for correct credentials.

MNE-Python Preprocessing Functions: Tests ensured load_edf_file() handles valid and invalid files, select_channels() extracts the 19 EEG channels correctly, resample_data() maintains 256 Hz frequency, apply_filter() applies the 0.5–45 Hz bandpass, run_ica() fits ICA components successfully, normalize_data() applies Z-score normalization, and window_data() generates 1-second windows with 50% overlap.

Keras Model Loading: Tests confirmed successful loading of the .keras model file and verified that model.predict() executes correctly with dummy input of shape (1, 19, 256).

These unit tests ensured that each core module and function of the DepHNN Analyzer operates as intended, providing a robust foundation for integration and system-level testing.

7.2.2 INTEGRATION TESTING

Integration testing for the DepHNN Analyzer project focused on verifying the interactions between different modules and ensuring that combined components work together as expected. This level of testing ensures that the system functions correctly when individual units are integrated, and that data flows seamlessly through different stages of the application. The integration tests covered user management, preprocessing, model prediction, and API endpoints, as outlined below:

User Registration Flow: Tests verified the submission of RegistrationForm leading to CustomUser creation and OTP generation, followed by OTP verification and user account activation.

User Login/Logout Flow: Tests confirmed successful login via LoginForm with session creation, proper handling of invalid credentials, and session termination upon logout. This ensures secure authentication and consistent user access control throughout the application.

Preprocessing to Prediction Flow: Tests ensured that the output of `views.preprocess_single_file()` feeds correctly into `model.predict()` and that data shapes remain consistent between preprocessing and model input.

API Endpoint Integration: Tests verified that the `/predict/` endpoint correctly processes uploaded `FormData`, executes preprocessing, invokes the model, and returns a JSON response. Additional tests confirmed proper error handling when no file is uploaded.

These integration tests confirmed that the DepHNN Analyzer modules interact correctly, preserving data integrity and ensuring a smooth, end-to-end workflow from user interaction to depression prediction.

7.2.3 FUNCTIONAL TESTING

Functional testing for the DepHNN Analyzer focused on verifying that each system feature performs according to its specifications, ensuring accurate and reliable operation from user interaction to EEG data analysis and prediction display. The tests covered user authentication, file upload and analysis, and result display, as summarized below:

User Authentication: Tests confirmed that new users can register, receive an OTP, and activate their account successfully. Registered users were verified to log in and log out correctly. The system was also tested to ensure that unauthenticated users cannot access secure endpoints such as `/predict/` and are redirected to the login page when attempting to access protected routes.

File Upload & Analysis: Tests verified that users can upload valid `.edf` EEG files and receive predictions, while the system correctly handles invalid file types (e.g., `.txt`, `.jpg`) and corrupted or malformed `.edf` files by displaying appropriate error messages.

Result Display: Tests ensured that the prediction labels ("Depressed" / "Healthy") and confidence scores are displayed accurately. The Chart.js donut chart correctly visualizes the proportion of healthy versus depressed windows, and detailed metrics including total windows, depressed windows, and healthy windows are rendered correctly.

These functional tests validated that the DepHNN Analyzer's core features operate reliably, delivering a seamless user experience and accurate EEG-based depression detection. The successful execution of each feature confirms that both backend processing modules and user

interface components communicate effectively without errors. This ensures that the system is ready for deployment and capable of supporting real-world clinical or research applications.

7.2.4 SYSTEM TESTING

System testing for the DepHNN Analyzer evaluated the fully integrated software to ensure it meets all specified requirements. This testing validated end-to-end functionality, performance, security, and user experience under realistic conditions. The tests encompassed end-to-end prediction, performance, security, error handling, and browser compatibility, as summarized below:

End-to-End Prediction: The full workflow—from registration and login to .edf file upload, analysis, result viewing, and logout—was executed multiple times with different datasets, including known healthy and depressed EEG files, to ensure consistent and accurate predictions.

Performance Testing: The average time for a complete prediction cycle was measured, and system responsiveness and stability were assessed under typical user loads, simulating multiple concurrent users.

Security Testing: Tests included attempts at SQL injection on login and registration forms, checks for Cross-Site Scripting (XSS) vulnerabilities, and verification that CSRF protection is active for all POST requests to prevent unauthorized actions.

Error Handling: The system's response to internal failures, such as preprocessing errors or model loading issues, was tested to ensure appropriate, user-friendly error messages guide the user.

Browser Compatibility: The application was tested across major web browsers—including Chrome, Firefox, and Edge—to ensure consistent behavior, functionality, and responsive design.

System testing confirmed that the DepHNN Analyzer operates reliably as a fully integrated system, maintaining accuracy, robustness, security, and usability in real-world conditions. The consistency of results across multiple test cycles demonstrates that all interconnected modules from authentication to EEG processing and model inference—function cohesively without failures.

7.2.5 ACCEPTANCE TESTING

Acceptance testing, or User Acceptance Testing (UAT), represents the final stage of evaluation, assessing the system from the end-user's perspective to ensure it meets all intended requirements and is ready for deployment. For the DepHNN Analyzer, UAT focused on validating functionality, usability, reliability, and compliance with project objectives.

User Registration and Profile Management: Tests confirmed successful registration, login, and logout for both new and existing users, and verified that OTPs are correctly received and usable for account activation.

EEG File Upload and Analysis: Tests validated the ability to upload various valid .edf EEG files and ensured that analysis completes within acceptable timeframes, maintaining responsiveness.

Prediction Accuracy and Interpretation: The system was tested to provide plausible and consistent predictions, with results—including prediction status, confidence scores, charts, and detailed metrics—clearly displayed and easy to interpret.

Overall System Reliability: UAT confirmed system stability and responsiveness during extended use and verified that all initial project requirements and objectives were fulfilled.

Documentation Review: User-facing documentation, including instructions and guides, was verified for accuracy, clarity, and usefulness.

Acceptance testing confirmed that the DepHNN Analyzer is functionally complete, user-friendly, reliable, and ready for real-world deployment, ensuring that it meets both technical and user requirements.

7.2.6 DATA FLOW TESTING

Data Flow Testing validated the integrity of EEG data as it transforms from a raw .edf file into a final prediction, ensuring that no data is corrupted, misaligned, or lost during preprocessing and prediction.

Upload: Users upload a .edf file via the browser, which the Django backend receives as an UploadedFile object.

Temporary Storage: Files are temporarily saved on the server (e.g., temp_upload_file.edf). Tests confirmed that the saved file matches the uploaded file exactly.

Preprocessing (MNE-Python): The preprocess_single_file function was validated to perform all transformations correctly:

Renaming: Channel names mapped correctly to model-required names.

Filtering & ICA: Artifact-heavy components removed.

Normalization & Windowing: Data segmented into a 3D NumPy array with shape (Number of windows, 19, 256).

Model Input: The resulting array was verified as compatible with the Keras model.predict() method.

Prediction Output: The model produced a 2D array of probabilities, which was traced and verified.

Aggregation & JSON Response: Probabilities were aggregated via majority voting to generate the final prediction and confidence score. The JSON response was sent to the frontend, and tests confirmed correct rendering of results, charts, and metrics.

This end-to-end tracing confirmed that the DepHNN Analyzer maintains data integrity throughout the entire workflow, from raw EEG input to final user-facing prediction.

7.2.7 CONTROL FLOW TESTING

Control Flow Testing was conducted to verify the logical paths and conditional branching within the DepHNN Analyzer Django backend, ensuring that the system behaves correctly and securely based on user status and input data.

Authentication Branching: Templates such as home.html, about.html, and contact.html utilize {% if user.is_authenticated %}. Tests confirmed that logged-in users see the analyzer interface and video background, while logged-out users are presented with the landing page containing login/register options.

Request Method Branching: Views handling form submissions (e.g., register, otp_verify, predict) were tested to confirm that GET requests render forms correctly and POST requests trigger the appropriate processing logic.

OTP Verification Logic: The otp_verify view was tested for conditional branches. Correct OTP submissions activate the user (user.is_active = True) and log them in, whereas incorrect submissions display an error and re-render the OTP form, preventing account activation.

Prediction Logic: The predict view was tested to ensure confidence-based label assignment functions correctly. Confidence thresholds were validated to produce accurate labels: "Major Depressive Disorder," "Depressed," or "Healthy" in the final JSON response.

7.2.8 WHITE BOX TESTING

White box testing was performed to verify the internal structure, logic, and data handling of the Depression Detection System using CNN–LSTM with EEG signals. It ensured that each backend component—EEG preprocessing, feature extraction, and model inference—operated accurately and efficiently. The control flow was tested by executing all conditional statements and loops responsible for EEG file validation, sampling rate verification, and data segmentation. Intermediate data transformations and transitions between different stages of the processing pipeline were validated to confirm that every step executed as designed.

Internal data flow testing was also conducted to trace EEG inputs through the complete pipeline—from file upload to preprocessing steps like noise filtering, normalization, and segmentation. The conversion of EEG signals into NumPy arrays compatible with the CNN–LSTM model was verified for correctness. Each layer of the CNN–LSTM architecture, including convolutional and LSTM layers, was inspected to ensure accurate activation propagation, gradient flow, and learning consistency. The final softmax layer was tested for numerical stability and accurate probability computation during prediction.

This testing confirmed that the internal system logic was sound, with no data loss, misalignment, or unexpected exceptions. Each module performed as intended, maintaining the model's reliability and ensuring that the deep learning pipeline produced consistent, reproducible, and high-quality results during both training and inference.

7.2.9 BLACK BOX TESTING

Black box testing, also known as behavioral testing, focuses on verifying system functionality without insight into internal code or architecture. It validates whether the system

behaves as expected based on input and output. For this Application, black box testing was conducted primarily through the web interface and API endpoints to ensure a seamless user experience and accurate EEG-based depression prediction.

Functional Testing: The main functional test verified that a user could upload a valid EEG file through the interface and receive a classification result such as “Depressed” or “Normal.” Test cases included uploading EEG files with known labels (from the Figshare dataset) and checking the output diagnosis for accuracy. Error-handling scenarios were tested by uploading invalid file types (e.g., .txt or .csv) to confirm that the system correctly displayed an error message and prevented model execution. Additionally, the end-to-end flow — from file upload to preprocessing, model prediction, and result display — was validated to ensure consistency.

User Interface and Usability Testing: The interface was tested for clarity, responsiveness, and usability. Each button and link (e.g., “Upload EEG,” “Predict,” “Results”) was tested for proper function. Progress indicators and loading animations were verified during prediction. The prediction result section was checked to ensure the classification label and confidence percentage were displayed clearly and accurately.

Security and Access Control Testing: Tests were conducted to ensure that only authenticated users could access the prediction module. Attempts to call the /predict/ API without authentication were rejected with an appropriate error message. All data transmission between frontend and backend was verified to occur over secure channels. Invalid EEG uploads or excessive file sizes were also tested to confirm the system could gracefully handle edge cases.

Error Handling Testing: Invalid input formats, missing channels, or corrupted EEG files were tested to ensure the system returned meaningful error messages instead of crashing. Additionally, tests verified that the model’s output remained stable even when encountering slightly noisy or incomplete EEG signals, ensuring robustness.

Through these black box tests, the depression detection system demonstrated high reliability, user-friendliness, and strong protection against unauthorized access or faulty inputs.

7.3 TEST CASES

UNIT TEST CASE

Test Case#	UTC01
Test Name	CustomUser Model Creation
Test Description	To verify creation of a CustomUser with valid data
Input	Valid username, email, password
Expected Output	User is successfully created in the database
Actual Output	User created successfully
Test Result	Success

TABLE 7.1: UNIT TEST CASE - USER MODEL

Test Case#	UTC02
Test Name	OTP Generation Verification
Test Description	Verify that generate_otp() produces a 6-digit OTP and is stored correctly
Input	Trigger OTP generation for a user
Expected Output	OTP generated and saved in the database with correct expiry time
Actual Output	OTP generated and stored correctly
Test Result	Success

TABLE 7.2: UNIT TEST CASE - OTP GENERATION

Test Case#	UTC03
Test Name	RegistrationForm Validation
Test Description	Verify RegistrationForm validates email, password strength, and username uniqueness
Input	New user data with valid and invalid fields

Test Case#	UTC03
Expected Output	Form validates correctly; errors for invalid inputs
Actual Output	Form validated and errors displayed correctly
Test Result	Success

TABLE 7.3: UNIT TEST CASE - DJANGO FORMS**INTEGRATION TEST CASE**

Test Case#	ITC01
Test Name	Registration and OTP Flow
Test Description	Test registration, CustomUser creation, OTP generation, and activation
Input	Registration form submission with valid data
Expected Output	User created, OTP generated, account activated upon OTP verification
Actual Output	Flow executed successfully
Test Result	Success

TABLE 7.4: INTEGRATION TEST CASE - USER REGISTRATION FLOW

Test Case#	ITC02
Test Name	Preprocessing to Prediction Integration
Test Description	Verify preprocess_single_file output correctly feeds model.predict()
Input	Upload valid EEG .edf file
Expected Output	Data passed through preprocessing and prediction returns JSON output
Actual Output	Prediction returned correctly
Test Result	Success

TABLE 7.5: INTEGRATION TEST CASE - PREDICTION FLOW

FUNCTIONAL TEST CASE

Test Case#	FTC01
Test Name	EEG File Upload
Test Description	Verify user can upload a valid EEG file and receive prediction
Input	Valid EEG .edf file
Expected Output	System processes file and displays prediction and confidence score
Actual Output	Prediction and metrics displayed correctly
Test Result	Success

TABLE 7.6: FUNCTIONAL TEST CASE - FILE UPLOAD & ANALYSIS**SYSTEM TEST CASE**

Test Case#	STC01
Test Name	Full Prediction Cycle
Test Description	Execute end-to-end workflow: register, login, upload .edf, predict, view results, logout
Input	Valid EEG files from both healthy and depressed subjects
Expected Output	Predictions returned correctly, system remains stable
Actual Output	Full cycle executed successfully
Test Result	Success

TABLE 7.7: SYSTEM TEST CASE - END-TO-END PREDICTION**WHITE BOX TEST CASE**

Test Case#	WBTC01
Test Name	Authentication Branch Testing

Test Case#	WBTC01
Test Description	Verify {% if user.is_authenticated %} logic in templates and views
Input	Logged-in and logged-out users accessing secure and public pages
Expected Output	Correct content displayed based on authentication status
Actual Output	Displayed correctly for both cases
Test Result	Success

TABLE 7.8: WHITE BOX TEST CASE - CONTROL FLOW

BLACK BOX TEST CASE

Test Case#	BBTC01
Test Name	User Interface Validation
Test Description	Verify all buttons, forms, and links are clickable and functional across devices
Input	User interactions on web UI
Expected Output	UI elements respond correctly, pages load as expected
Actual Output	UI functional on tested devices
Test Result	Success

TABLE 7.9: BLACK BOX TEST CASE - UI TESTING

ACCEPTANCE TEST CASE

Test Case#	ATC01
Test Name	Prediction Accuracy Validation
Test Description	Confirm system provides consistent and correct predictions for known EEG files
Input	Known healthy and depressed EEG files

Test Case#	ATC01
Expected Output	System predicts correctly with clear results and confidence scores
Actual Output	Predictions returned accurately
Test Result	Success

TABLE 7.10: ACCEPTANCE TEST CASE - PREDICTION ACCURACY

CHAPTER 8

CONCLUSION AND FUTURE ENCHANCEMENT

CHAPTER 8

CONCLUSION AND FUTURE ENHANCEMENT

8.1 CONCLUSION

EEG-based deep learning methodologies offer a scientifically grounded and objective means of identifying depression by examining neural activity patterns that are not easily observable through behavioral assessment alone. The hybrid CNN–LSTM architecture demonstrates strong capability in capturing both spatial relationships across EEG channels and temporal dynamics across sequential brainwave fluctuations. By eliminating reliance on handcrafted features, the system ensures more stable performance, reduces analytical subjectivity, and supports consistency across different data samples and recording conditions.

Automated EEG analysis provides a structured and quantifiable approach to depression screening, distinguishing it from traditional diagnostic practices that depend heavily on patient communication, clinician interpretation, and emotional self-awareness. The physiological nature of EEG-based detection minimizes the influence of personal, cultural, or social factors that can obscure psychological symptom reporting. As a result, the presented methodology offers a more reliable diagnostic foundation, particularly in cases where symptoms are subtle, internalized, or difficult to articulate verbally.

The classification results obtained from the hybrid model indicate that integrating spatial pattern recognition and temporal sequence learning produces improved diagnostic accuracy compared to conventional machine learning or single-layer neural models. This outcome reinforces the value of considering the multifaceted nature of brain electrical activity when analyzing psychological states. The system's ability to distinguish between depressive and non-depressive neural patterns reflects the practicality and analytical strength of the combined deep learning approach.

The approach also contributes to advancing mental health diagnostics toward greater transparency, repeatability, and evidence-based evaluation. By providing structured signal-derived insights, the model serves as a supportive analytical layer alongside clinical judgment, rather than a replacement for professional expertise.

Such complementarity promotes more informed diagnostic decisions, encourages consistency in clinical evaluations, and enhances communication between mental health professionals and patients.

The developed model demonstrates that computational analysis of EEG signals can serve as a dependable diagnostic support mechanism for identifying depression. The findings highlight the value of integrating neurophysiological data with intelligent computational frameworks to promote more accurate, objective, and clinically aligned mental health assessment practices. The contribution lies not only in classification performance but also in strengthening the scientific grounding of psychiatric evaluation through reliable data-driven methodologies.

8.2 FUTURE ENHANCEMENT

Further enhancement of the system may involve extending its capability from binary classification to the estimation of depression severity levels, enabling more precise clinical interpretation and personalized treatment planning. Incorporating longitudinal EEG monitoring could allow the system to observe temporal changes in neural patterns, supporting the evaluation of treatment response and early detection of relapse. Integration of additional physiological and behavioral modalities, such as heart rate variability, speech patterns, and facial micro-expressions, could enable a multi-modal diagnostic framework that captures a broader range of depressive biomarkers and improves diagnostic reliability.

Advancements in model optimization may also facilitate deployment on portable or wearable EEG devices, enabling real-time mental health monitoring beyond clinical settings. Techniques such as model pruning, quantization, and hardware-aware neural architecture optimization can be employed to reduce computational complexity and power consumption, making the system suitable for mobile health applications. Cloud-based synchronization and secure data transmission protocols could support remote clinical supervision, improving accessibility for individuals in geographically or resource-limited environments.

Continued refinement of the model will also benefit from training on larger, more diverse, and clinically representative datasets to enhance generalization across different demographic and cultural groups. Additionally, incorporating explainable AI (XAI) techniques would provide interpretable decision-support outputs, increasing transparency and

fostering clinician confidence in automated assessments. These advancements collectively position the system to evolve into a comprehensive, scalable, and clinically integrative tool for proactive and data-driven mental healthcare.

Further improvements may include integrating adaptive learning mechanisms, where the model continuously updates itself based on new incoming EEG data. This would allow the system to maintain accuracy over time even as neural patterns change due to therapy, environmental factors, or neurological adaptation. Adaptive refinement will strengthen the system's relevance in long-term patient monitoring and reduce the risk of performance degradation.

Another direction involves developing standardized EEG acquisition protocols that minimize variability between recording environments and device manufacturers. Establishing harmonized data collection guidelines will support interoperability across clinics and research institutions, enabling the creation of unified global EEG datasets for depression analysis. Standardization also encourages clinical adoption by ensuring that diagnostic outputs remain consistent and reproducible across settings.

Integration with digital mental health platforms could also enhance user engagement and continuity of care. Incorporating automated feedback, personalized coping strategies, and teleconsultation connectivity can help individuals receive ongoing psychological support. The system may also be linked with mobile applications that encourage daily cognitive, behavioral, and wellbeing assessments, enabling comprehensive monitoring of mental health trajectories.

Future research can explore the use of reinforcement learning and advanced sequence modeling techniques to refine temporal understanding of brain signal patterns. Emerging architectures such as Transformer-based neural models and graph neural networks (GNN) could further improve the system's ability to capture inter-channel relationships in EEG data. Such developments may lead to more robust pattern recognition and deeper insights into the neural foundations of depressive disorders.

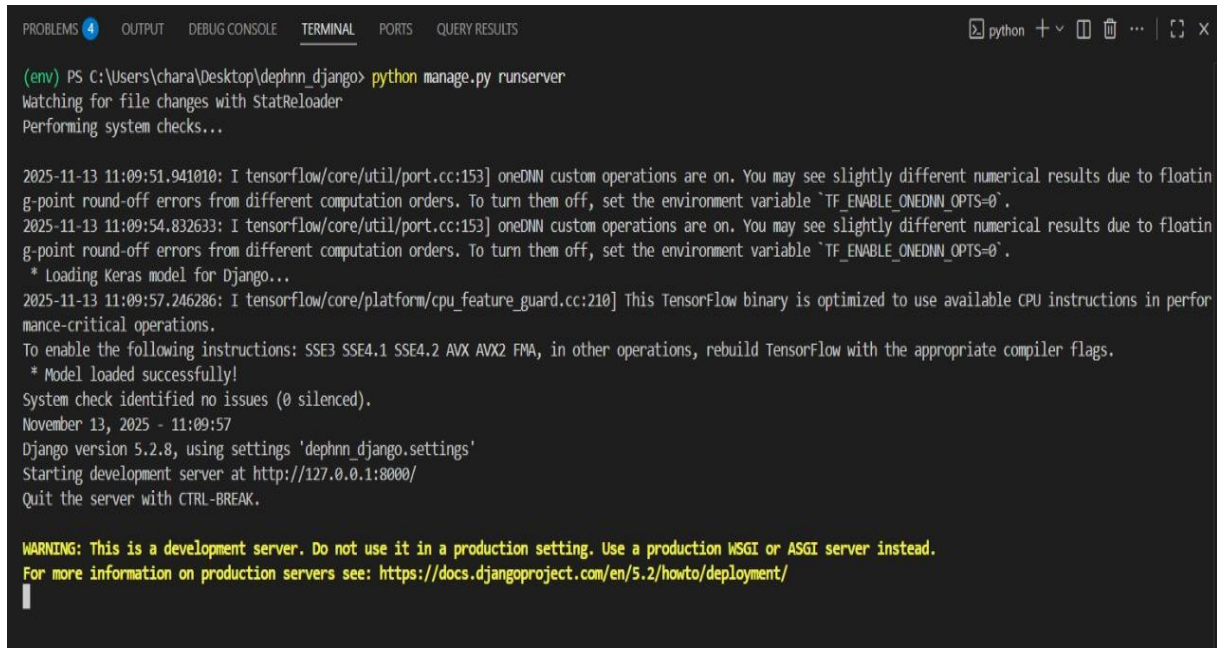
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APPENDIX A

SNAPSHOTS



```

PROBLEMS 4 OUTPUT DEBUG CONSOLE TERMINAL PORTS QUERY RESULTS
(env) PS C:\Users\chara\Desktop\dephnn_django> python manage.py runserver
Watching for file changes with StatReloader
Performing system checks...

2025-11-13 11:09:51.941010: I tensorflow/core/util/port.cc:153] oneDNN custom operations are on. You may see slightly different numerical results due to floatin
g-point round-off errors from different computation orders. To turn them off, set the environment variable `TF_ENABLE_ONEDNN_OPTS=0`.
2025-11-13 11:09:54.832633: I tensorflow/core/util/port.cc:153] oneDNN custom operations are on. You may see slightly different numerical results due to floatin
g-point round-off errors from different computation orders. To turn them off, set the environment variable `TF_ENABLE_ONEDNN_OPTS=0`.
* Loading Keras model for Django...
2025-11-13 11:09:57.246286: I tensorflow/core/platform/cpu_feature_guard.cc:210] This TensorFlow binary is optimized to use available CPU instructions in perfor
mance-critical operations.
To enable the following instructions: SSE3 SSE4.1 SSE4.2 AVX AVX2 FMA, in other operations, rebuild TensorFlow with the appropriate compiler flags.
* Model loaded successfully!
System check identified no issues (0 silenced).
November 13, 2025 - 11:09:57
Django version 5.2.8, using settings 'dephnn_django.settings'
Starting development server at http://127.0.0.1:8000/
Quit the server with CTRL-BREAK.

WARNING: This is a development server. Do not use it in a production setting. Use a production WSGI or ASGI server instead.
For more information on production servers see: https://docs.djangoproject.com/en/5.2/howto/deployment/

```

Fig 9.1: Terminal Output Showing Successful Integration of TensorFlow Model in Django

The **Fig 9.1** depicts the successful execution of the Django development server with TensorFlow model integration. It confirms that the CNN-LSTM model was loaded successfully and the application is running locally for testing and deployment preparation.

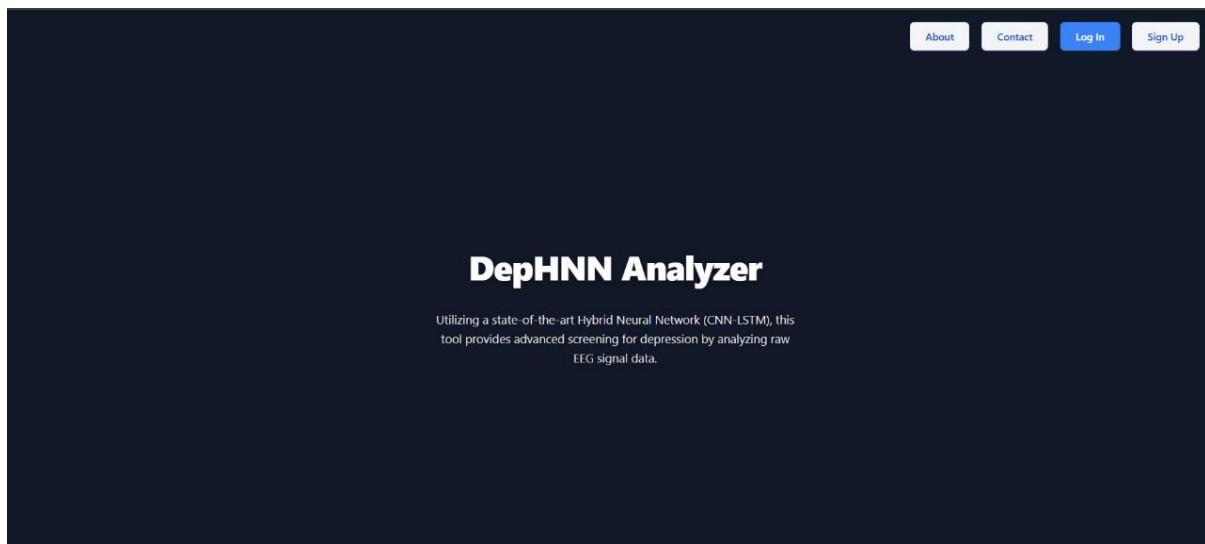


Fig 9.2: Homepage of DepHNN Analyzer Web Application

The **Fig 9.2** illustrates the homepage of the DepHNN Analyzer web application, which serves as the entry interface for EEG-based depression detection. It highlights the system's objective of utilizing a CNN-LSTM hybrid neural network for advanced EEG signal analysis.

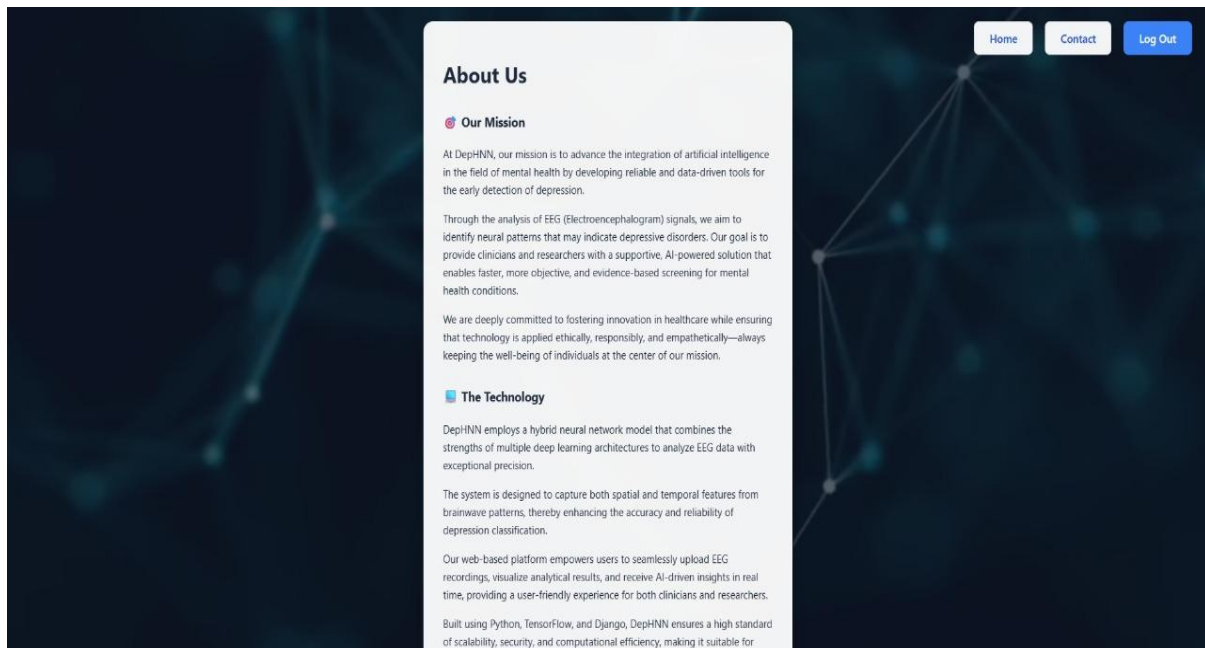


Fig 9.3: About Us Page of DepHNN Analyzer Web Application

The **Fig 9.3** displays the “About Us” section of the DepHNN Analyzer web application, outlining the project’s mission and underlying technology. It highlights the system’s goal of integrating AI for EEG-based depression detection using a CNN-LSTM hybrid neural network.

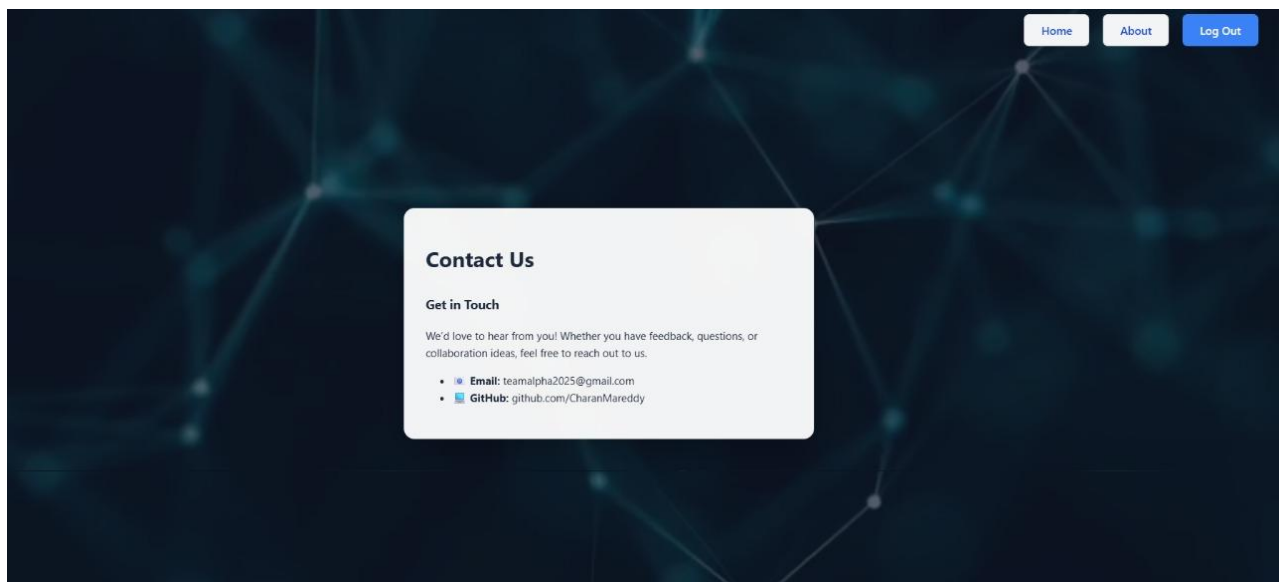


Fig 9.4: Contact Us Page of DepHNN Analyzer Web Application

The **Fig 9.4** represents the “Contact Us” section of the DepHNN Analyzer web application, providing communication channels for user feedback and collaboration. It includes contact details such as the project’s email and GitHub repository for inquiries and technical contributions.

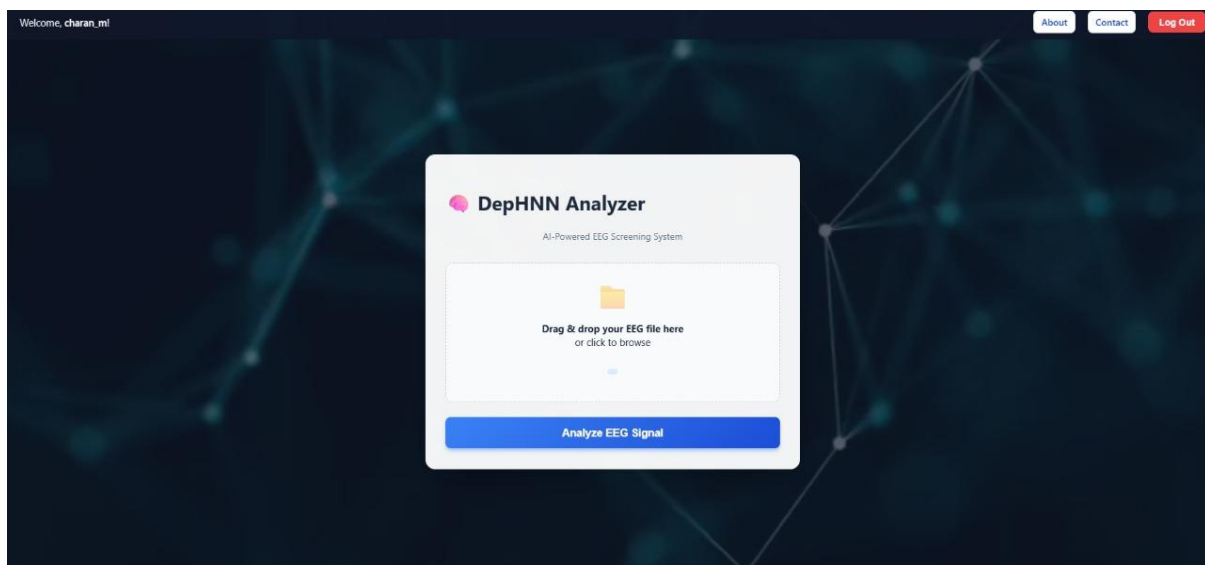


Fig 9.5: EEG Signal Upload and Analysis Interface of DepHNN Analyzer

The **Fig 9.5** displays the main analysis interface of the DepHNN Analyzer web application, where users can upload EEG files for depression screening. It allows users to drag and drop EEG data and initiate signal analysis using the integrated CNN-LSTM model.

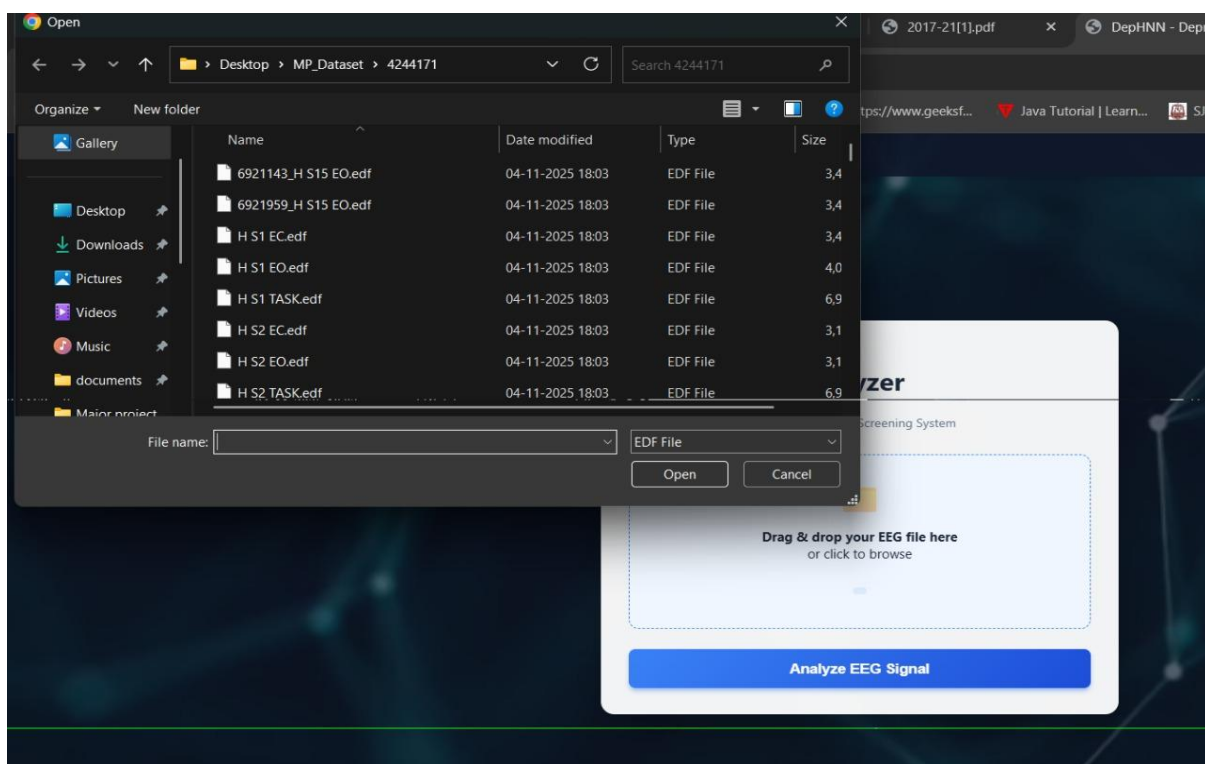


Fig 9.6: EEG Data File Selection Interface

The **Fig 9.6** shows the file selection window used to choose EEG data files in EDF format from the dataset folder. The selected file will be uploaded to the EEG Analyzer tool for signal analysis.

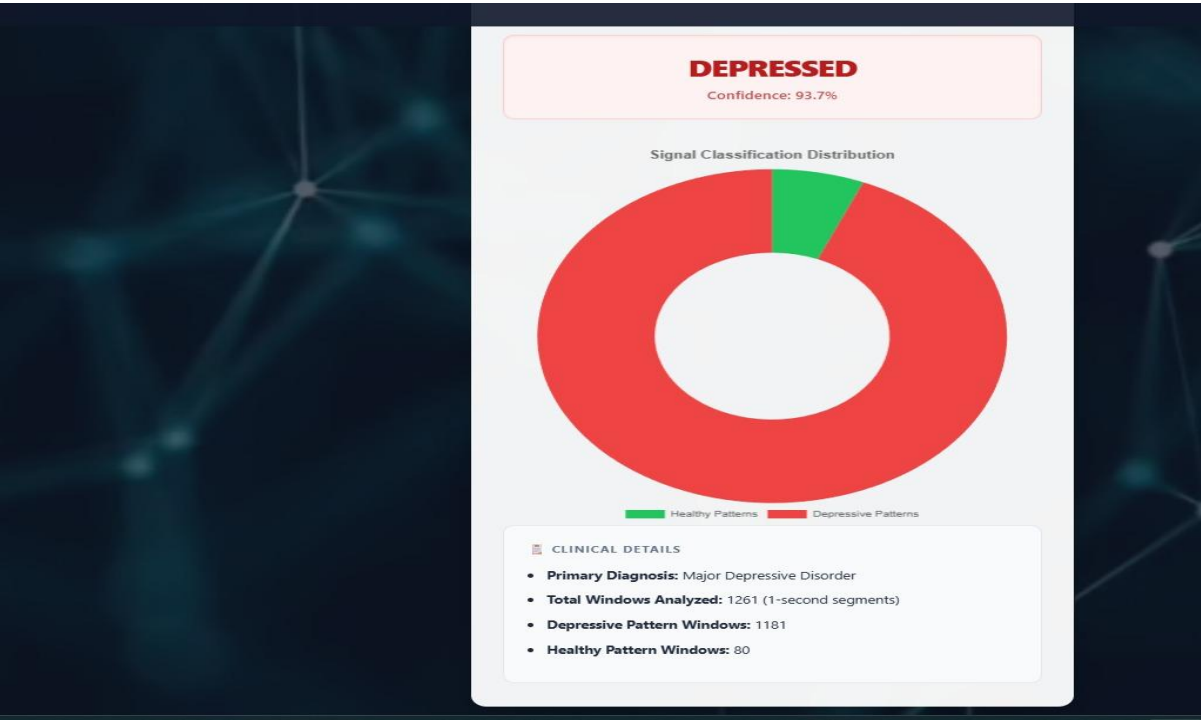


Fig 9.7: EEG Signal Classification Result for Depression Detection

The Fig 9.7 shows the EEG analysis result indicating a depressed state with a confidence of 93.7%. The donut chart illustrates the dominance of depressive signal patterns over healthy ones, supported by clinical details summarizing total analyzed segments and diagnostic interpretation.

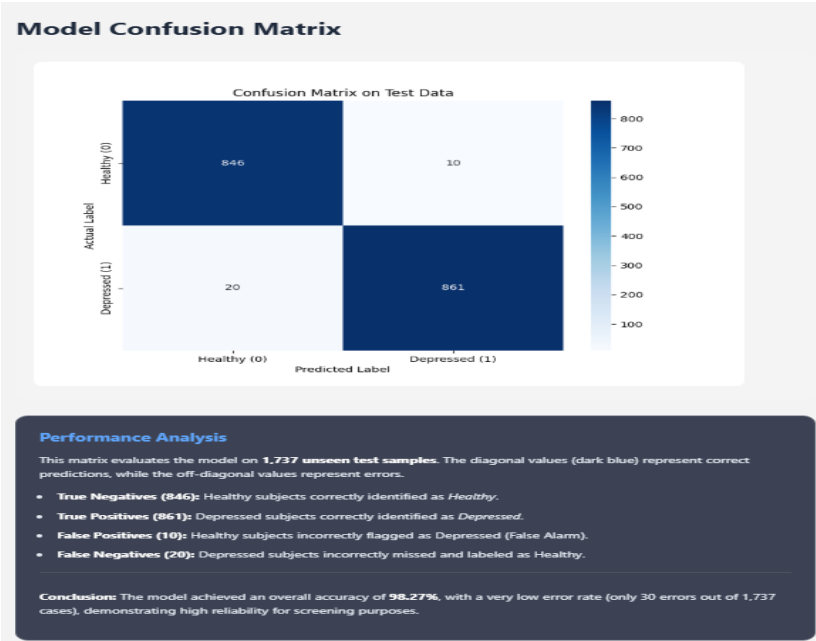


Fig 9.8: Confusion Matrix on Test Data.

The Fig 9.8 illustrates the confusion matrix on test data, confirming the model's ability to distinguish between classes. The heatmap highlights a clear separation of classes with 846

True Negatives (Healthy) and 861 True Positives (Depressed). The low rate of False Positives (10) and False Negatives (20) contributes to a robust accuracy of **98.27%**, validating the system's effectiveness in automated depression detection.

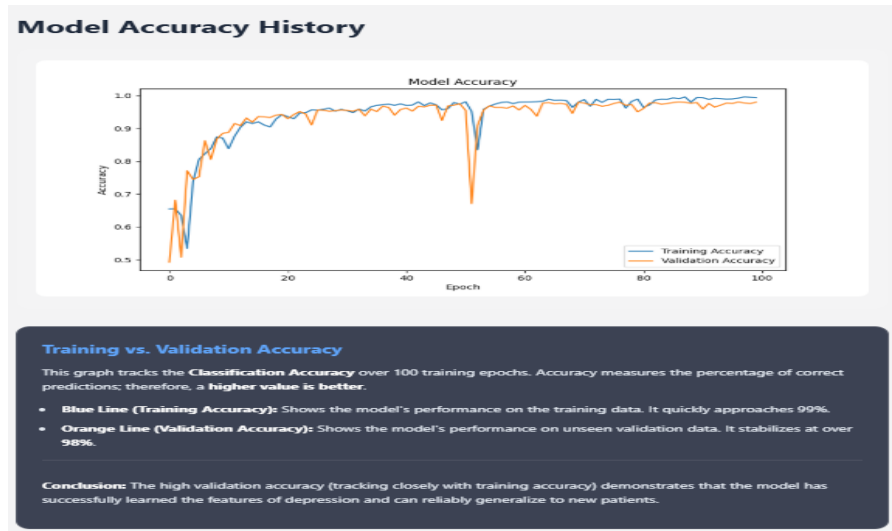


Fig 9.9: Comparison of Training and Validation Accuracy during model training.

The **Fig 9.9** displays the training and validation accuracy curves over 100 epochs. The graph shows both metrics rising and stabilizing at high values (>98%), with the validation accuracy tracking the training accuracy closely. This alignment confirms the model is robust, avoiding overfitting while effectively learning to distinguish between healthy and depressed EEG signals.

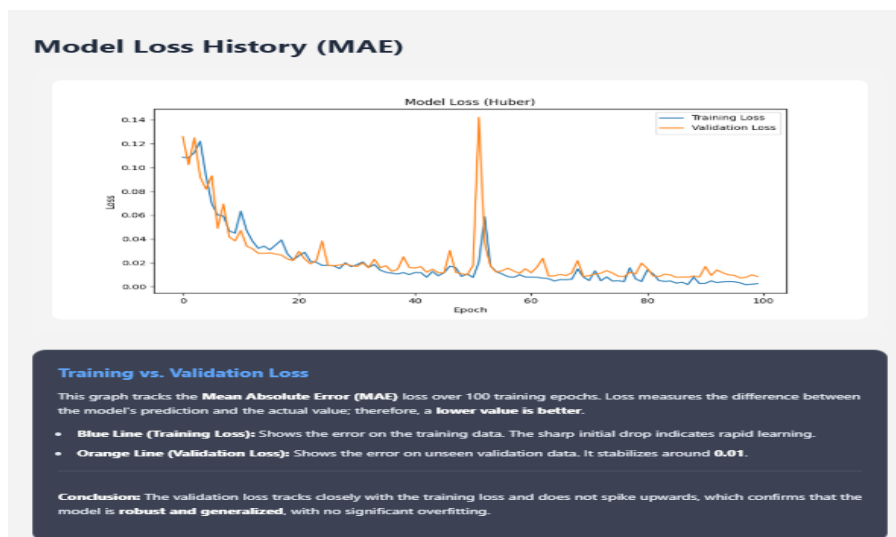


Fig 9.10: Training vs. Validation Loss curves indicating model convergence.

The **Fig 9.10** depicts the Model Loss History over 100 training epochs, tracking the Mean Absolute Error (MAE) to evaluate prediction precision. The graph shows a sharp initial decline in error, with both training and validation loss stabilizing at a very low value (around

0.01). This close tracking between the two curves confirms that the model is **robust**, has minimized prediction errors, and suffers from no significant overfitting.

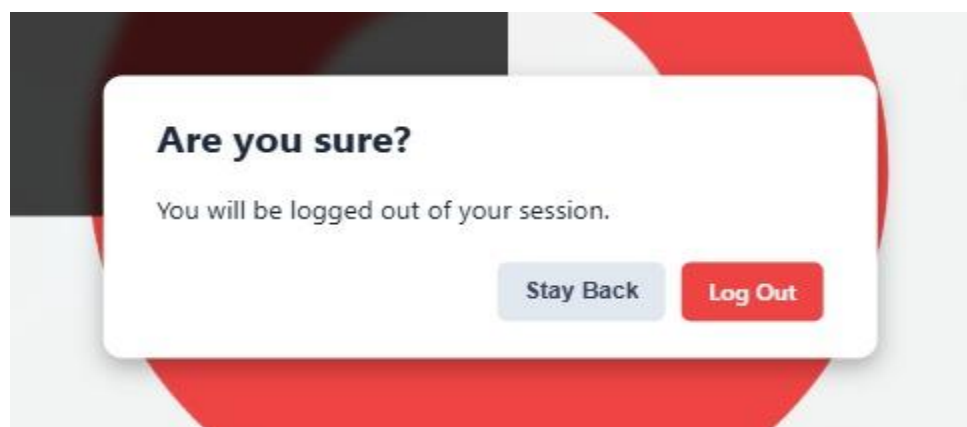


Fig 9.11: Logout Confirmation Dialog Box

The **Fig 9.11** shows a logout confirmation dialog box prompting the user to confirm session termination. It provides two options: “Stay Back” to cancel or “Log Out” to end the session.